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Restoration of locomotor function following stimulation of the A13 region in Parkinson's mouse models

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Abstract

Parkinson's disease (PD) is characterized by extensive motor and non-motor dysfunction, including gait disturbance, which is difficult to treat effectively. This study explores the therapeutic potential of targeting the A13 region, a dopamine-containing area of the medial zona incerta (mZI) that has shown relative preservation in PD models. The A13 is identified to project to the mesencephalic locomotor region (MLR), with a subpopulation of cells displaying activity correlating to movement speed, suggesting its potential involvement in locomotor function. We show that photoactivation of this region can alleviate bradykinesia and akinetic symptoms in a mouse model of PD, revealing the presence of preserved parallel motor pathways for movement. We identified areas of preservation and plasticity within the mZI connectome using whole-brain imaging. Our findings suggest a global remodeling of afferent and efferent projections of the A13 region, highlighting the zona incerta's role as a crucial hub for the rapid selection of motor function. Despite endogenous compensatory mechanisms proving insufficient to overcome locomotor deficits in PD, our data demonstrate that photostimulation of the A13 region effectively restores locomotor activity. The study unveils the significant pro-locomotor effects of the A13 region and suggests its promising potential as a therapeutic target for PD-related gait dysfunction.

Significance statement

This work examines the function of the A13 nucleus in locomotion, an area with direct connectivity to locomotor regions in the brainstem. Our work shows that A13 stimulation can restore locomotor function and improve bradykinesia symptoms in a PD mouse model.

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This **useful** study summarises the effect of optical stimulation of the A13 region on locomotion in healthy mice and experimental Parkinsonism and could potentially be of interest to basic and clinical neuroscientists. Behavioural analyses and evidence for pro-locomotor effects of stimulation are **solid**. However, anatomical analyses are **incomplete** and do not yield mechanistic insights due to various issues with specificity, sample size, statistical analysis, and data presentation in the present form of the study.

Introduction

Parkinson's disease (PD) is a complex condition affecting many facets of motor and non-motor functions, including visual, olfactory, memory and executive functions (Cenci and Björklund, 2020). Due to the widespread features of PD, focusing on changes within a single pathway cannot account for all symptoms. Gait disturbance is one of the hardest to treat; pharmacological, deep brain stimulation (DBS) and physical therapies lead to only partial improvements (Nonnekes et al., 2020, 2015). While the subthalamic nucleus (STN) and globus pallidus (GPi) are common DBS targets for PD, alternative targets such as pedunculopontine nucleus (PPN) and the zona incerta (ZI) have been proposed with mixed results in improving postural and/or gait dysfunctions (Caire et al., 2013; Ferraye et al., 2010; Gut and Winn, 2015; Hamani et al., 2011; Moro et al., 2010; Nonnekes et al., 2015; Okun and Foote, 2010; Ossowska, 2019; Stefani et al., 2007; Thevathasan et al., 2018). Part of the issue with targeting the ZI with DBS strategies is the relative lack of knowledge regarding its downstream anatomical and functional connectivity with motor centres. Recent work with photoactivation of subpopulations of PPN neurons in PD models shows promise for similar ZI-focused strategies (Masini and Kiehn, 2022).

The ZI is recognized as an integrative hub, with roles in regulating sensory inflow, arousal, motor function, and conveying motivational states (Mitrofanis, 2005; Wang et al., 2020). As such, it is well placed to be involved in PD and has seen increased clinical and preclinical research over the last two decades (Blomstedt et al., 2018; Ossowska, 2019; Plaha et al., 2008). However, little attention has been placed on the medial zona incerta (mZI), particularly the A13, the only dopamine-containing region of the rostral ZI (Bolton et al., 2015; Kim et al., 2017; Sharma et al., 2018). Recent research in primates and mice (Peoples et al., 2012; Roostalu et al., 2019; Shaw et al., 2010) indicates that the A13 is preserved in 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)-based PD models.

Recently, we discovered that the A13 located within the ZI projects to two areas of the mesencephalic locomotor region (MLR), the PPN and cuneiform nucleus (CnF)(Sharma et al., 2018), suggesting a role for A13 in locomotor function. Indeed, *in vivo* photometry recordings from calcium/calmodulin-dependent protein kinase IIα (CaMKIIα) populations in the rostral ZI, which includes the A13 nucleus, show a subpopulation of cells whose activity correlates with movement speed (Li et al., 2021). Since this region projects to the MLR, it is a potential parallel motor pathway to target for gait improvement. Photoactivation of glutamatergic MLR neurons alleviates motor deficits in the 6-OHDA mouse model (Fougère et al., 2021; Masini and Kiehn, 2022). Phenomena such as kinesia paradoxa (Glickstein and Stein, 1991) in PD patients support the existence of preserved parallel motor pathways that can be engaged in particular circumstances to produce normal movement.

Further evidence supporting the importance of parallel motor pathways in PD includes those reporting functional alterations in A13 (Hoffman et al., 1997; Périer et al., 2000). Nigrostriatal lesions affect A13 cellular function and lead to anatomical remodeling in monoaminergic brain regions (Braak et al., 2003; Kish et al., 2008; Lim et al., 2009; Perez-Lloret and Barrantes, 2016; Roostalu et al., 2019; Scatton et al., 1983; Zweig et al., 1989). The A13 connectome encompasses the cerebral cortex (Mitrofanis and Mikuletic, 1999), central nucleus of the amygdala (Eaton et al., 1994), thalamic paraventricular nucleus (Li et al., 2014), thalamic reuniens (Sita et al., 2007; Venkataraman et al., 2021), MLR(Sharma et al., 2018), superior colliculus (SC) (Bolton et al., 2015), and dorsolateral periaqueductal grey (PAG) (Messanvi et al., 2013; Sita et al., 2007), making the A13 an important hub for goal-directed locomotion (Choi and McNally, 2017; Eaton et al., 1994; Messanvi et al., 2013; Mok and Mogenson, 1986; Moriya et al., 2020; Ogundele et al., 2017; Manjit K. Sanghera et al., 1991; M. K. Sanghera et al., 1991; Sita et al., 2007; Venkataraman et al., 2021).

Based on the role of the A13 in gait and, specifically, as a possible target to improve gait in PD, we investigated the therapeutic potential of photoactivating a tightly circumscribed region targeting a small region containing mainly the A13 and a small area of the mZI, which we term A13 region throughout the manuscript. We identified areas of preservation and plasticity within the mZI connectome using whole-brain imaging techniques. Photoactivation of the A13 region rescued bradykinetic and akinetic symptoms in a mouse model of 6-hydroxydopamine (6-OHDA) mediated unilateral nigrostriatal degeneration. Because the zona incerta is a hub for the rapid selection of motor function, we then mapped the input and output patterns of the region. We found evidence of a global remodeling of afferent and efferent projections of the A13 region. While endogenous compensatory mechanisms from the remaining but remodelled A13 region connectome were inadequate in overcoming locomotor deficits observed in 6-OHDA mice, photostimulation of the A13 region restored locomotor activity. These data demonstrate that the A13 region produces powerful pro-locomotor effects in normal and PD mouse models. Moreover, PD-related bradykinesia is ameliorated with A13 region photoactivation in the presence of remodelling of the A13 region connectome. Some of these data have been published in abstract form (L. Kim et al., 2021).

Results

Unilateral 6-OHDA mouse model has robust motor deficits

The overall experimental design is illustrated in **Figure 1A**, along with a schematic in **Figure 1B** showing injections of 6-OHDA in the medial forebrain bundle and AAVDJ-CaMKIIα-ChR2 virus into the medial zona incerta (mZI). We confirmed substantia nigra pars compacta (SNc) degeneration in a well-validated unilateral 6-OHDA-mediated Parkinsonian mouse model (Thiele et al., 2012). The percentage of tyrosine hydroxylase (TH^+) cell loss normalized to the intra-animal contralesional side was quantified. 6-OHDA produced a significant lesion that decreased TH^+ neuronal SNc populations. As previously reported (Boix et al., 2015), the SNc ipilesional to the 6-OHDA injection (n = 10) showed major ablation of the TH^+ neurons compared to sham animals (**Figure 1C and D**: n = 11).

A13 region photoactivation generates pro-locomotor behaviors in the open field

6-OHDA lesions are characterized as generating bradykinetic and akinetic phenotypes in the open field (Li et al., 2022; Magno et al., 2019; Masini and Kiehn, 2022; Sanders and Jaeger, 2016). To understand the impact of A13 region photoactivation on locomotion in sham and PD model mice, on-target localization of ferrule above the A13 region, centered on the mZI, along with

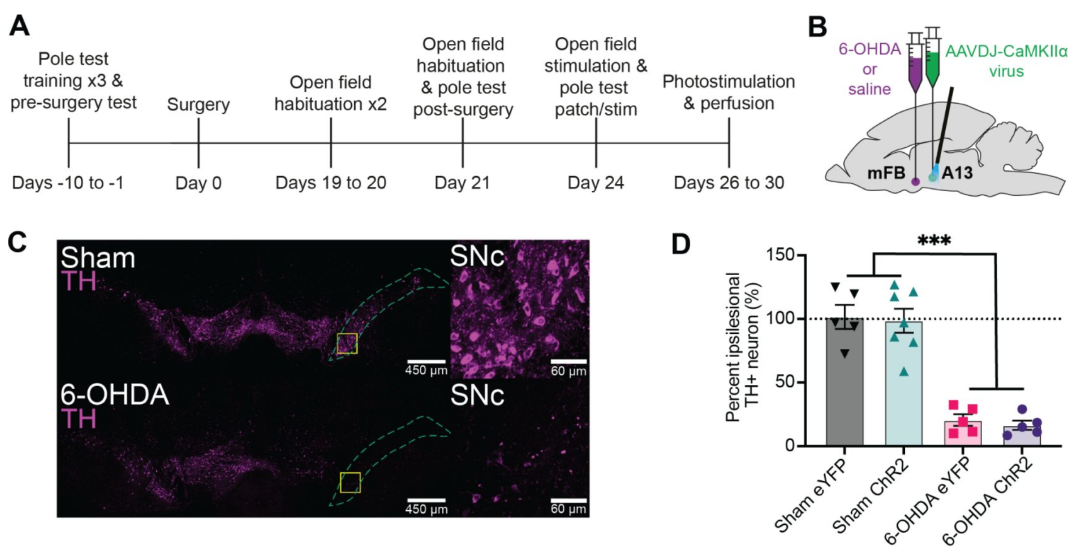

Figure 1.

Experimental design and confirmation of unilateral TH^+ depletion in the SNc via 6-OHDA lesion.

(**A**) Illustration of experimental timeline. (**B**) Dual ipsilateral stereotaxic injection into the MFB and A13 region. (**C**) TH^+ cells in SNc of sham (top) compared to 6-OHDA injected mouse (bottom). Magnified areas outlined by yellow squares are shown on the right. (**D**) Unilateral injection of 6-OHDA (6-OHDA ChR2: n = 5, 6-OHDA eYFP: n = 5) into the MFB resulted in greater percentage of TH^+ loss compared to sham in the SNc (sham ChR2: n = 7, sham eYFP: n = 5, three-way MM ANOVAs), regardless of virus type ($F_{1,18}$ = 104.4, p < .001). ***p < .001. Error bars indicate SEMs.

YFP reporter expression, was confirmed (**Figure 2**) in mice given sham or 6-OHDA injections. Corroborating the *post hoc* targeting, we found evidence for c-Fos in neurons within the A13 region in photostimulated ChR2 mice (**Figure 2**). Before *post hoc* analysis, mice were monitored in the open field test (OFT), where the effects of the 6-OHDA lesion were apparent, with 6-OHDA lesioned animals demonstrating far less movement, fewer bouts of locomotion, and less time engaging in locomotion in the OFT (**Figure 2A-E**). Notably, photoactivation of the A13 region often generated dramatic effects, with mice showing a distinct increase in locomotor behavior (**Figure 3A**, Movie S1 & Movie S2). Both sham and 6-OHDA ChR2 mice showed a significant increase in locomotor distance travelled during periods of photoactivation (**Figure 3B**, p = 0.005). One sham animal showed grooming behavior on stimulation and was excluded from the analysis.

We tested whether photoactivation led to a single bout of locomotion or if there was an overall increase in bouts, signifying that animals could repeatedly initiate locomotion following photoactivation. Mice in the ChR2 groups demonstrated an increase in the number of locomotion bouts with photoactivation, indicating a greater ability to start locomotion from rest, and that photoactivation was not eliciting a single prolonged bout (**Figure 3C**, p = 0.005). When we examined each bout of locomotion, photoactivation increased the total duration of locomotion (**Figure 3D**, p = 0.005). There was a refractory decrease in the distance travelled by the sham ChR2 animal group (**Figure S1A**), which was not evident for the 6-OHDA cohort (**Figure S1B**). To control for this, we compared the pre-timepoints to the baseline one-minute averages to ensure that the animal locomotion distance travelled returned to a stable state before stimulation was reapplied (**Figure S1C**, p = 0.783).

Next, we examined the reliability of photoactivation to initiate locomotion. The percentage of trials with at least one bout of locomotion was compared for the pre-and stim time points. 6-OHDA ChR2 animals showed a reliable pro-locomotion phenotype with A13 region photoactivation (**Figure S2A**: p = 0.042). As was expected in the control 6-OHDA eYFP group, there was no effect of photoactivation on the probability of engaging in locomotion (p = 0.713).

Animal movement speed also factors into the total distance travelled measure and can be discussed in regard to a bradykinetic phenotype in 6-OHDA lesioned mice (Magno et al., 2019; Masini and Kiehn, 2022; Sanders and Jaeger, 2016). Using instantaneous animal movement speeds that exceeded 2 cm/s as per Masini & Kiehn (2022), we plotted instantaneous speed (**Figure 3F-I**) and analyzed one-minute bins (**Figure 3E**). As was expected, 6-OHDA lesioned animals had lower movement speeds than sham control animals (p < 0.001). One animal from the 6-OHDA eYFP group was excluded because it did not meet the speed threshold during recording. Both the 6-OHDA ChR2 and sham ChR2 groups displayed increases in average speed during photostimulation (**Figure 3E**, p < 0.001). When we examined the time taken to initiate locomotion, there was no significant difference between sham or 6-OHDA ChR2 groups (**Figure S2B**).

Photoactivation of the A13 region increases ipsilesional turning in the open field test

Unilateral 6-OHDA lesions drive asymmetric rotational bias (Boix et al., 2015; Li et al., 2022; Magno et al., 2019; Thiele et al., 2012). We were interested in whether this persisted with stimulation and noted upon observation of photoactivation that animals appeared to have increased ipsilesional rotation. As observed, 6-OHDA ChR2 animals had an increase in turn angle sum (TAS), indicating an increase in their rotational bias with photoactivation in the ipsilesional direction (**Figure 3J**, p < 0.001). As expected, 6-OHDA eYFP animals showed consistent rotational bias throughout time. The rotational bias of sham ChR2 was also compared to determine whether the increased rotational bias was due to photoactivation or the interaction of photoactivation and the lesion. The sham ChR2 group showed no significant change in TAS with photoactivation

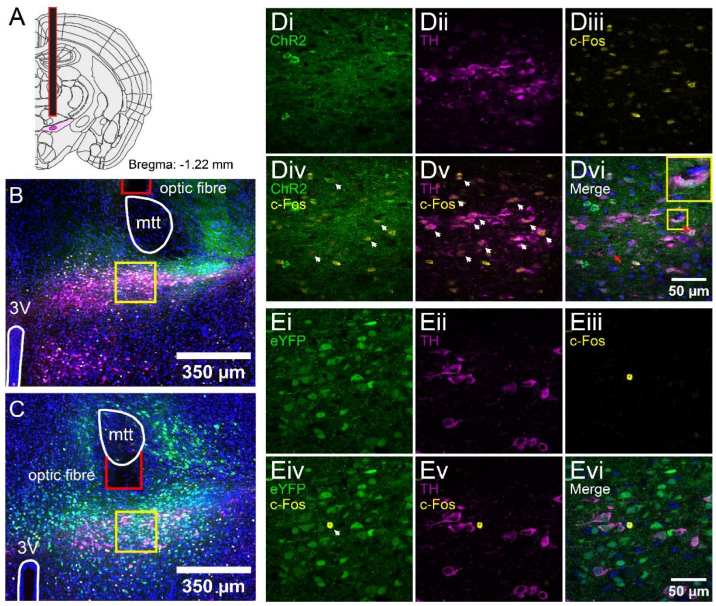

Figure 2.

Post hoc **c-Fos expression and targeting of the mZI and A13.**

(A) Diagram showing the A13 DAergic nucleus in dark magenta encapsulated by the ZI in light magenta. The fibre optic tip is outlined in red. Atlas image adapted from the Allen Brain Atlas (Goldowitz, 2010). **(B)** Tissue images were obtained from 6-OHDA ChR2 animals around bregma −1.22 mm, and **(C)** a 6-OHDA eYFP animal more caudally around bregma −1.46 mm. Images show the distribution of DAPI (blue), eYFP (green), c-Fos (yellow), and TH (magenta). Landmarks are outlined in white (3V: third ventricle; mtt: mammillothalamic tract), and the optic cannula tip is shown in red. Higher magnification images of the A13 DAergic nucleus are outlined by the yellow boxes in a 6-OHDA ChR2 animal **(D)** and a 6-OHDA eYFP animal **(E)**. Scale bars are set to 350 µm. Images show isolated channels in the top rows of the respective groups: eYFP **(i)**, TH **(ii)**, and c-Fos **(iii)**. Merged channels for eYFP and c-Fos **(iv)**, TH and c-Fos **(v)**, and a merge of all four channels **(vi)** are presented in the bottom rows of their respective groups. White arrowheads in the merged images highlight overlap in merged markers. Red arrows show triple colocalization of eYFP, c-Fos and TH. **(Dvi)** contains a magnified example of triple-labelled neurons, as highlighted in the yellow box. Scale bars are set to 50 µm.

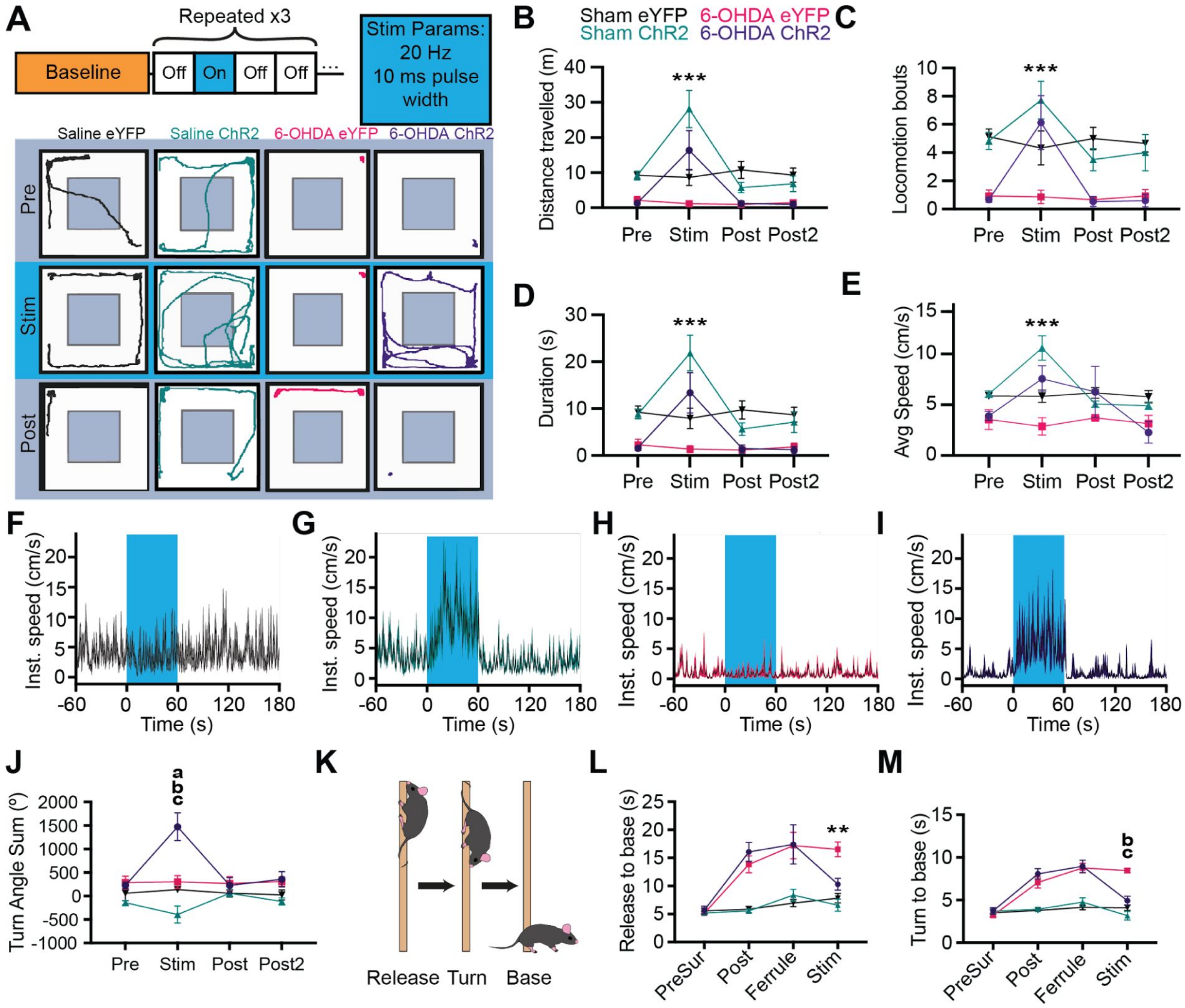

Figure 3.

Ispilesional photoactivation of the A13 region in a unilateral 6-OHDA mouse model rescues motor deficits.

(**A**) Schematic of open field experiment design and example traces for open field testing (1 min) with unilateral photoactivation of the A13 region. (**B-E**) Effects of photoactivation on open field metrics for sham eYFP (n = 5), sham ChR2 (n = 6), 6-OHDA eYFP (n = 5), and 6-OHDA ChR2 (n = 5) groups (three-way MM ANOVAs, *post hoc* Bonferroni pairwise). Photoactivation increased in the ChR2 groups: (**B**) distance travelled (ChR2 vs. eYFP: p = 0.005), (**C**) locomotor bouts (ChR2 vs. eYFP: p = 0.005), (**D**) duration of locomotion in the open field (ChR2 vs. eYFP: p = 0.005), and (**E**) animal movement speed (ChR2 vs. eYFP: p < 0.001). (**F-I**) Group averaged instantaneous velocity graphs showing no increase in a sham eYFP (**F**) or 6-OHDA eYFP mouse (**H**), with increases in velocity during stimulation in a sham ChR2 (**G**) and 6-OHDA ChR2 (**I**) mouse. (**J**) The graph presents animal rotational bias using the turn angle sum. There was a significant increase in 6-OHDA ChR2 rotational bias during A13 region photoactivation (6-OHDA ChR2 vs. 6-OHDA eYFP: p < 0.001). (**K**) Diagram depicting the pole test. A mouse is placed on a vertical pole facing upwards. The time for release is taken as the experimenter removes their hand from the animal's tail. (**L, M**) Graphs showing the response of animals to photoactivation of the A13 region while performing the pole test. (**L**) A13 region photoactivation also led to shorter total descent time in ChR2 compared to eYFP mice (ChR2 vs. eYFP: p = 0.004), and (**M**) 6-OHDA ChR2 mice showed a greater reduction in descent time compared to sham ChR2 (6-OHDA ChR2 vs. sham ChR2: p = 0.012; 6-OHDA ChR2: n = 5; sham ChR2: n = 7). ***p < .001, **p < .01, *p < .05. Bonferroni's *post hoc* comparisons between 6-OHDA ChR2 and sham eYFP, sham ChR2, and 6-OHDA eYFP at stim time point as a, b, and c respectively. Error bars indicate SEMs.

(**Figure 3J**, $p > 0.05$). Next, we examined whether the increased turning angle sum in the 6-OHDA ChR2 group was observed during periods of locomotion. When the TAS was calculated only during periods of locomotion, the rotational bias in the animal orientation in the 6-OHDA ChR2 animals was not observed ($p = 0.286$).

Skilled vertical locomotion is improved in the pole test with photoactivation of the A13 region

The pole test is a classic 6-OHDA behavioral paradigm (**Figure 3K**) that involves skilled locomotor abilities for an animal to turn and descend a vertical pole (Matsuura et al., 1997; Ogawa et al., 1985). Improvements in function can be inferred if the time taken to complete the test decreases (Matsuura et al., 1997; Ogawa et al., 1985). 6-OHDA mice demonstrated significantly greater descent times than sham mice ($p < 0.001$). Photoactivation of the A13 region reduced descent times for both 6-OHDA and sham groups on the pole test (**Figure 3L**, $p = 0.004$, Movie S3). Neither of the eYFP groups showed any changes in the time to complete the pole test.

To further understand the effects of photoactivation on the ability of mice to descend the pole, the time taken for mice to descend after turning was analyzed to remove any influence of animals spending time investigating their environment on the top of the pole. While all groups showed reduced total pole test descent time with photoactivation, considering just the time to descend from turn alone, there was a larger improvement with A13 region photoactivation in the 6-OHDA ChR2 mice compared to sham ChR2 mice (**Figure 3M**: $p = 0.012$). These results indicate that photoactivation has the effect of reducing bradykinesia by improving the ability of mice to descend the pole during the PT.

Dopaminergic Cells in the A13 region are preserved in the unilateral 6-OHDA mouse model

While photoactivation of the A13 region promoted locomotor activity in both sham and 6-OHDA mice, there were differences in speed and directional bias. We hypothesized that this may be due to changes in the A13 region connectome since there is evidence of changes in firing and metabolic activity in the region (Périer et al., 2000). Therefore, we utilized whole brain imaging approaches (Hansen et al., 2020; Zhan et al., 2021) to examine changes in the connectome following 6-OHDA lesions of the nigrostriatal region.

Using whole brain imaging, as expected, TH^+ cells in SNc were more vulnerable to the 6-OHDA neurotoxin than the ventral tegmental area (VTA) and A13 (**Figure S3A-F**). 6-OHDA-treated mice showed a significantly greater percentage of TH^+ cell loss in SNc compared to the VTA and A13 (VTA vs. SNc: $p = 0.003$; A13 vs. SNc: $p = 0.005$). In contrast, sham animals showed no significant difference in TH^+ cell loss across SNc, VTA and A13 (**Figure S3G**, $p > 0.05$). Thus, similar to that observed in the human brain of Parkinsonian patients (Matzuk and Saper, 1985), there is a remarkable preservation of dopaminergic cells in the A13 after nigrostriatal degeneration in the 6-OHDA mouse model of PD.

Large-scale changes in the A13 region connectome following 6-OHDA-mediated unilateral nigrostriatal degeneration

Although photoactivation had benefits in restoring speed in 6-OHDA mice (**Figure 3**), circling behavior was increased, suggesting additional changes that may reflect connectome alterations. We examined the changes in the input and output of the A13 by co-injecting anterograde (AAV8-CamKII-mCherry) and retrograde AAV (AAVrg-CAG-GFP) tracers into the A13 nucleus (Keith B. J. Franklin and Paxinos, 2008). The injection core and spread were determined in the rostrocaudal direction from the injection site (**Figure S4**). To examine whether unilateral nigrostriatal degeneration resulted in changes in the organization of inputs and outputs from the A13, we first visualized interregional correlations of afferent and efferent proportions for each condition using correlation matrices (**Fig 4A and B**; 251 regions in a pairwise manner). Correlation matrices were organized using the hierarchical anatomical groups from the Allen Brain Atlas (**Figure 4C**). To minimize the influence of experimental variation on the total labeling of neurons and fibers, the afferent cell counts or efferent fiber areas in each brain region were divided by the total number found in a brain to obtain the proportion of total inputs and outputs. The data were normalized to a $\log_{10}$ value to reduce variability and bring brain regions with high and low proportions of cells and fibers to a similar scale (Kimbrough et al., 2020). Comparing the afferent and efferent proportions in a pairwise manner between mice showed good consistency with an average correlation of 0.91 ± 0.02 (Spearman's correlation, **Figure S5**).

We observed changes in projection patterns between the sham and the 6-OHDA group. A correlation matrix was used to quantify the relationship of input from brain regions in a pairwise manner through the neuraxis. Overall, afferents onto the A13 in sham animals displayed a higher interregional correlation between brain regions than 6-OHDA-injected mice. Specifically, correlation coefficients of 0.10, 0.30, and 0.50 or larger represent weak, moderate, and strong correlations, respectively (Cohen, 1988). Compared to sham (**Figure 4C**), in 6-OHDA injected mice (**Figure 4D**), afferent contributions from two clusters of brain subregions became more dissimilar (anti-correlated) to the rest of the neuraxis: 1) cortical plate subregions (motor, sensory, visual, and prefrontal), and 2) striatal, and pallidal subregions compared to sham (boxed blue areas in **Figure 4D**). These data suggest that afferents from several regions showed a coordinated reduction in afferent density onto the A13, except contributions from cortical plate (motor, sensory, visual, and prefrontal), striatal, and pallidal subregions that were positively correlated. In other words, contributions from the cortical plate (motor, sensory, visual, and prefrontal), striatal and pallidal subregions are positively correlated, but compared to the rest of the brain, they are anti-correlated. This suggests a greater afferent input onto A13 from the cortical plate (motor, sensory, visual, and prefrontal), striatal and pallidal subregions than other regions after 6-OHDA lesions.

In marked contrast, the projection patterns of A13 efferents exhibited a higher level of interregional correlation between brain regions following a unilateral nigrostriatal degeneration compared to sham. In the sham condition, the A13 connectome is biased towards cortical and striatal regions compared to pallidal, thalamic, hypothalamic, midbrain efferents. This is shown by a broad negative correlation between these two large groups (**Figure 4E**). However, these broad anti-correlations disappear following nigrostriatal degeneration (**Figure 4F**). These data indicate that the A13 efferent connectome is less refined following nigrostriatal degeneration.

Differential remodeling of A13 region connectome ipsi- and contra-lesion following 6-OHDA-mediated nigrostriatal degeneration

The distributions of the A13 connectome in sham animals served as the basis for an in-depth comparison of the preservation and plasticity of A13 afferents and efferents in 6-OHDA mouse models (**Figure 5**). We observed a global remodeling of A13 afferents and efferents following unilateral nigrostriatal degeneration (**Figure 5A**, B; E, F) that was differentially expressed across the neuraxis (**Figure 4D**, H). The ipsilesional side showed more downregulated areas (**Figure 5D**: see also example traces). These downregulated areas were focused within the cortical plate and cortical subplate regions. 6-OHDA injections also downregulated A13 afferent densities from the striatum, pallidum, thalamus and medulla. While 6-OHDA mainly downregulated ipsilesional A13 afferent densities, the hypothalamus (including ZI), midbrain, pons, and cerebellum had increased A13 afferent densities.

The intact, contralesional side showed more upregulated regions across the neuraxis, suggesting compensatory upregulation in the unilateral 6-OHDA model (**Figure 5D**). The cortical plate, striatum, and cortical subplate were the top three upregulated contralesional regions. In contrast,

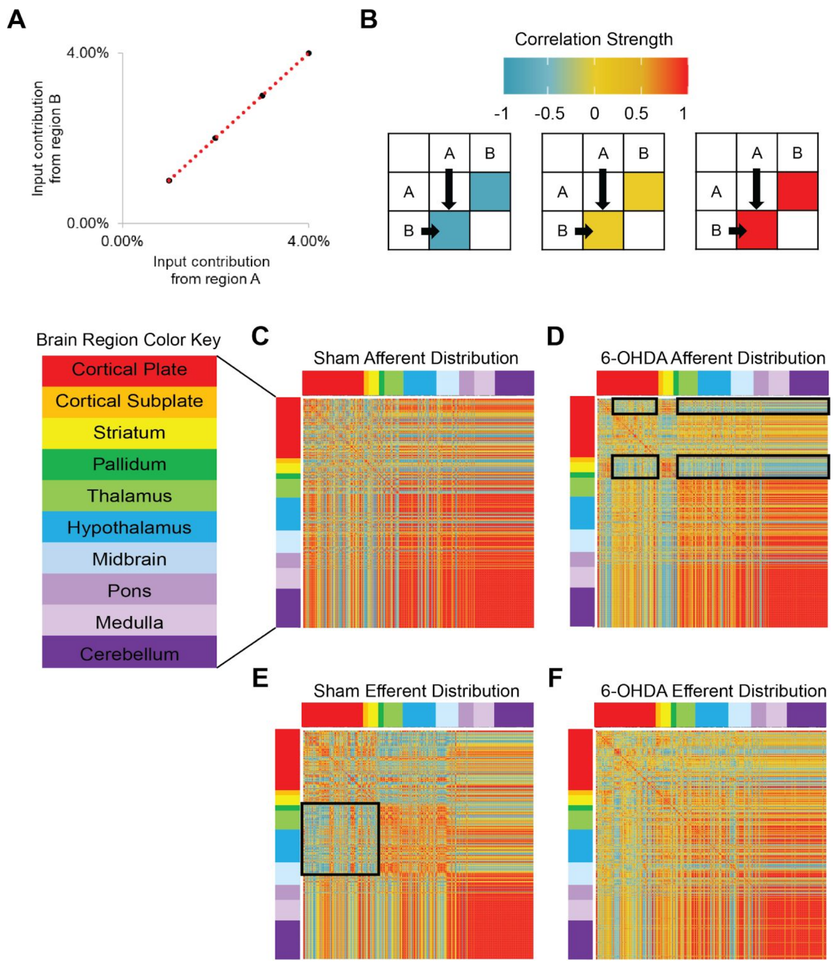

Figure 4.

Unilateral nigrostriatal degeneration leads to large-scale changes in the organization of the A13 region afferent and efferent distributions across the neuraxis.

We used correlation matrices to summarize any observable patterns in the distribution patterns of inputs and outputs of the A13 region. A correlation matrix was calculated by correlating the proportion of input from one brain region to another in a pairwise manner across 251 brain regions delimited by registration with Allen Brain Atlas. If two brain regions among mice (eg. brain regions A and B) contribute a similar input, they are highly correlated **(A)**. Using the color legend showing various correlation strengths, the intersecting box in the matrix in this example will be colored dark red **(B)**. If no relationship is found between contributions from two brain regions, the intersecting box will be colored yellow. If the contribution from one brain region was negatively correlated with another brain region among mice, then the intersecting box will be colored blue. The afferent distribution pattern in the sham displayed a higher level of inter-regional correlation between brain regions **(C)** than 6-OHDA injected mice **(D)**. Indeed, two distinct bands of anti-correlated afferent regions were identified in the 6-OHDA injected mice (see black boxes in **D**). These two bands arose from the cortical plate subregions (motor, sensory, visual, and prefrontal), and striatal and pallidal subregions showing distinct inputs compared to the rest of the neuraxis. In contrast, the projection patterns of A13 efferents displayed a higher level of inter-regional correlation between brain regions following a unilateral nigrostriatal degeneration **(F)** compared to sham **(E)**. In sham, proportions of A13-cortical/ striatal efferents were negatively correlated to A13-pallidal/ thalamic/ hypothalamic/ midbrain efferents (see black boxes in **E**). However, these distinct projection patterns disappeared following nigrostriatal degeneration, suggesting A13 efferent distributions becoming more distributed across the neuraxis.

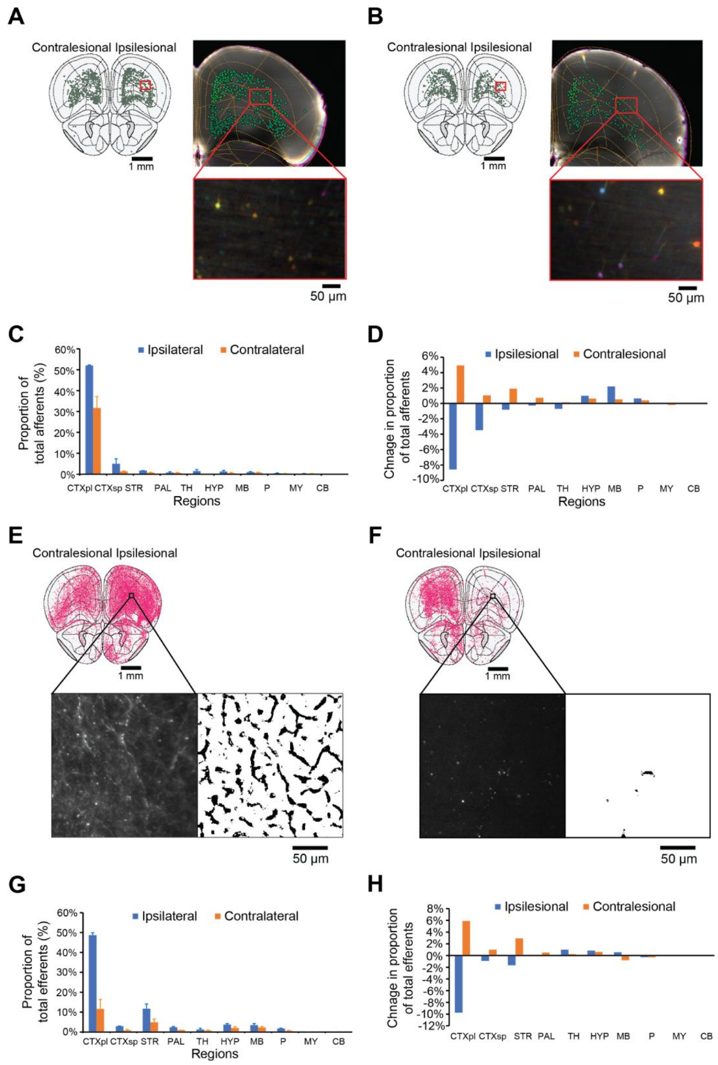

Figure 5.

Differential remodeling of A13 region connectome following a unilateral nigrostriatal degeneration.

The distributions of the A13 connectome in sham served as a basis for an in-depth comparison against 6-OHDA mouse models. Example registered slices (using WholeBrain software 64 with light-sheet data, 2X objective, 4X optical zoom) at rostral areas show changes in sham **(A)** afferents compared to 6-OHDA lesioned animals **(B)**. Graph showing major brain regions contributing afferents to A13 in sham mice **(C)**. The graph illustrates the change in the proportion of afferents in 6-OHDA compared to sham mice **(D)**. Representative registered slices showing sham proportions of efferents in sham **(E)** compared to 6-OHDA mice **(F)**. The magnified black box section displays an example of mCherry+ fibers (left) segmented using Ilastik and ImageJ software (right). Graph showing major brain regions receiving efferents from the A13 in sham mice **(G)**. The graph illustrates the change in the proportion of efferents in 6-OHDA compared to sham mice **(H)**. Error bars represent SEMs. Anterograde and retrograde viruses were injected into the ipsilesional A13 (see methods). Abbreviations from Allen Brain. Atlas: CTXpl (cortical plate), CTXsp (cortical subplate), STR (striatum), PAL (palladium), TH (thalamus), HYP (hypothalamus), P (pons), MB (midbrain), MY (Medulla), and CB (cerebellum).

contralesional A13 afferent densities from the pallidum and thalamus were spared and upregulated. Thus, compensatory upregulation of A13 afferent density from these regions appeared lateralized from the intact, contralesional side. Furthermore, bilateral compensatory upregulation of A13 afferents was observed from the hypothalamus (including ZI), midbrain and pons.

The A13 efferents were more downregulated on the ipsilesional side (**Figure 5H**). However, the downregulation was focused within the isocortical, striatal and cortical subplate regions. Remodeling on the contralesional efferent projection patterns closely followed the changes seen with the afferents, except for projections onto the thalamic and midbrain regions. The A13 efferents onto thalamic regions were bilaterally upregulated. Also, the A13-midbrain efferents were upregulated ipsilesionally.

Discussion

Our work demonstrates robust pro-locomotor effects induced by photoactivation of the A13 region in lesioned and sham mice. Photoactivation during the OFT increased locomotion distance travelled, the duration of locomotion, and speed in both sham and 6-OHDA mice. Uniquely, the 6-OHDA group had increases in the number of locomotor bouts which resembled the normal number of bouts observed in healthy mice at baseline. Bradykinesia in 6-OHDA mice was substantially improved following photoactivation. We found extensive input and output connectivity of the A13, which was remodeled following nigrostriatal lesions. Afferent input patterns displayed a marked reduction in interregional correlation across brain regions in 6-OHDA mice, while efferent projections increased. This demonstrates the impact of nigrostriatal lesions on dopaminergic-containing regions outside the nigrostriatal zone. These findings highlight the pro-locomotory effect, therapeutic potential, and plasticity of the A13.

The role of the A13 region in locomotion in sham mice

We provide the first direct evidence of the photoactivation of the A13 being sufficient in driving locomotion. It is now evident that the pro-locomotor function of ZI extends further rostrally than previous work in caudal ZI (cZI) indicates (Mitrofanis, 2005): photoactivation of cZI neurons increased animal movement speed in prey capture (Zhao et al., 2019) and active avoidance (Hormigo et al., 2020). Previous data targeting the mZI region, including somatostatin (SOM^+), calretinin (CR^+), and $vGlut2^+$ neurons, did not change locomotor distance travelled in the OFT (Li et al., 2021). In our work, there may be a combinatorial effect of multiple populations being photostimulated or targeting more medial populations in the ZI. Our results are consistent with *in vivo* calcium dynamics from $CaMKII\alpha^+$ rostral ZI cells, which overlap the A13 showing subpopulations whose activity correlates with either movement speed or anxiety-related locations (Li et al., 2021). Enhancement of the A13 activity appears to modulate locomotor activity in naive mice differently from more lateral GABAergic ZI populations (dorsal and ventral ZI). Microinjection of $GABA_A$ receptor agonists, muscimol (Wardas et al., 1988) or etomidate (Chen et al., 2023), into the ZI either evokes severe catalepsy or a significant reduction in locomotor distance and velocity, respectively. Suppression of GABAergic ZI activity can either increase locomotion by microinjection of $GABA_A$ receptor antagonist bicuculline (Périer et al., 2002) or induce bradykinesia and akinesia by chemogenetic or optogenetic inhibitions in healthy naive mice (Chen et al., 2023).

Furthermore, our previous work showing A13 projections to the MLR is consistent with our observed photoactivation effects. We, and others, demonstrated that the A13 contains DA neurons, which may contribute to the observed effects, possibly via $D_{1/5}$ receptor activation (Ryczko et al., 2016). A13 region photoactivation produces increased locomotor speed averaging 13 cm/s and improves descent times on the pole test. The enhanced ability to perform the pole test, which requires the animal to grasp the vertical pole to descend safely without falling, provides further evidence for the role of A13 region neurons in movements. Since A13 stimulation did not alter coordination during the task, it suggests a complex behavioral role consistent with its upstream location from the brainstem and extensive connectome. A13 photoactivation increases animal speed, duration, and distance travelled. Interestingly, the latency in A13 to observe increases in ongoing animal speed or to initiate locomotion is long in both 6-OHDA and sham mice (~ 15 and 5 seconds, respectively). Second-long delays are often typical of sites upstream of the cuneiform, such as the dlPAG, which has a delay of several seconds (Tsang et al., 2021). The delays in locomotor initiation and context-specific integration following stimulation of upstream CnF targets may offer a therapeutic advantage for overcoming gait dysfunction.

Photoactivation of the A13 reduces bradykinesia and akinesia in mouse PD models

Several studies have focused on the basal ganglia by targeting the subthalamic nucleus (Gradinaru et al., 2009; Yoon et al., 2014), endopeduncular nucleus (Moon et al., 2018; Yoon et al., 2020), SNc (Kravitz et al., 2010), striatum (Bordia et al., 2016; Ryan et al., 2018), cZI (Li et al., 2022), and motor cortex (Magno et al., 2019; Sanders and Jaeger, 2016; Valverde et al., 2020), or their projections in mouse models of PD. MLR subpopulations have been explored as a target for PD DBS with mixed results (Fougère et al., 2021; Masini and Kiehn, 2022). Recently, Li et al. found that cZI glutamatergic neurons were overactive after administering 6-OHDA into the striatum, and photoinhibition rescued the motor deficits (Li et al., 2022). Motor deficits in a 6-OHDA-induced PD mouse model were also ameliorated by chemogenetic and optogenetic activation of dorsal and ventral ZI GABAergic neurons (Chen et al., 2023). Here, we introduce a novel subpopulation in ZI (A13 cells) whose photoactivation rescued bradykinetic and akinetic deficits observed in the 6-OHDA mice.

Our work shows that A13 projections are affected at cortical and striatal levels following 6-OHDA, consistent with our observed changes in locomotor function. Over 28 days, there was a remarkable change in the afferent and efferent A13 regional connectome, despite the preservation of TH^+ ZI cells. This is consistent with previous reports of widespread connectivity of the ZI (Mitrofanis, 2005). The preservation of A13 is expected since A13 lacks DAT expression (Bolton et al., 2015; Negishi et al., 2020; Sharma et al., 2018) and is spared from DAT-mediated toxicity of 6-OHDA (Dauer and Przedborski, 2003; Konnova et al., 2018; Simola et al., 2007). While A13 cells are spared following nigrostriatal degeneration, our work demonstrates its connectome is rewired. The ipsilateral afferent projections were markedly downregulated, while contralesional projecting afferents showed upregulation. In contrast, efferent projections showed less downregulation in the cortical subplate regions and bilateral upregulation of thalamic and hypothalamic efferents. Similar timeframes for anatomical and functional plasticity affecting neurons and astrocytes following an SNc or MFB 6-OHDA have been previously reported (Bosson et al., 2015; Perović et al., 2005; Requejo et al., 2020). Human PD brains that show degeneration of the SNc have a preserved A13 region, suggesting that our model, from this perspective, is externally valid (Matzuk and Saper, 1985).

Combined with photoactivation of the A13 region, we provide evidence for plasticity following damage to SNc. A previous brain-wide quantification of TH levels in the MPTP mouse model identified additional complexity in regulating central TH expression compared to conventional histological studies (Roostalu et al., 2019). Roostalu et al. reported decreased SNc TH^+ cell numbers without a significant change in TH^+ intensity in SNc and increased TH^+ intensity in limbic regions such as the amygdala and hypothalamus (Roostalu et al., 2019). Likewise, we found no significant change in A13 TH^+ cell counts. Still, there was a downstream shift in the distribution pattern of A13 efferents following nigrostriatal degeneration with a pullback on outputs to cortical and striatal subregions. This suggests A13 efferents are more distributed across the neuraxis than in sham mice. The preserved A13 efferents could provide compensatory dopaminergic innervation with collateralization mediated contralesionally and, in some subregions, ipsilesionally to increase the availability of extracellular dopamine. Considering A13-MLR efferents (Sharma et al., 2018) that remain preserved, photoactivation of glutamatergic MLR neurons alleviates motor deficits in the 6-OHDA mouse model (Fougère et al., 2021; Masini and Kiehn, 2022), and photoactivation of A13 somata promotes locomotion in 6-OHDA mice - hypotheses that warrant further investigation.

Several A13 efferent targets could be responsible for rotational asymmetry. In a unilateral 6-OHDA model, ipsiversive circling behaviour is indicative of intact striatal function on the contralesional side (Carey, 1991; Schwarting et al., 1991; Ungerstedt, 1971; Zetterström et al., 1986). Instead, the predictive value of a treatment is determined by contraversive circling mediated by increased dopamine receptor sensitivity on the ipsilesional striatal terminals (Costall et al., 1976; Lane et al., 2006). Thus, our data suggest that photoactivation of ipsilesional A13 has an overall additive effect on ipsiversive circling and represents a gain of function on the intact side that contributes to the magnitude of overall motor asymmetry against the lesioned side. Since A13 cells are preserved in PD, future therapies could use bilateral stimulations optimized for each side to minimize the overall motor asymmetry while ameliorating bradykinesia and akinesia.

With the induction of a 6-OHDA lesion, there is a change in the A13 connectome, characterized by a reduction in bidirectional connectivity with ipsilesional cortical regions. In rodent models, the motor cortices, including the M1 and M2 regions, can shape rotational asymmetry (Gradinaru et al., 2009; Magno et al., 2019; Sanders and Jaeger, 2016; Valverde et al., 2020). Activation of M1 glutamatergic neurons increases the rotational bias (Valverde et al., 2020), while M2 neuronal stimulation promotes contraversive rotations (Magno et al., 2019). Our data suggest that A13 photoactivation may have resulted in the inhibition of glutamatergic neurons in the contralesional M1. An alternative possibility is the activation of the contralesional M2 glutamatergic neurons, which would be expected to induce increased ipsilesional rotations (Magno et al., 2019). The ZI could generate rotational bias by A13 modulation of cZI glutamatergic neurons via incerto-incertal fibres (Ossowska, 2019; Power and Mitrofanis, 1999), which promotes asymmetries by activating the SNr (Li et al., 2022). The incerto-incertal interconnectivity has not been well studied, but the ZI has a large degree of interconnectivity (Sharma et al., 2018; Tsang et al., 2021) along all axes and between hemispheres (Power and Mitrofanis, 1999). However, this may only contribute minimally given that unilateral photoactivation of the A13 cells in sham mice failed to produce ipsiversive turning behavior while unilateral photoactivation of cZI glutamatergic neurons in sham animals was sufficient in generating ipsiversive turning behavior (Li et al., 2022). Another possibility involves the A13 region projections to the MLR. With the unknown downstream effects of A13 photoactivation, there may be modulation of the PPN neurons responsible for this turning behavior (Masini and Kiehn, 2022). The thigmotaxic behaviors suggest some effects may be mediated through dlPAG and CnF (Tsang et al., 2021), and recent work suggests the CnF as a possible therapeutic target (Fougère et al., 2021; Noga and Whelan, 2022). Since PD is a heterogeneous disease, our data provide another therapeutic target providing context-dependent relief from symptoms. This is important since PD severity, symptoms, and progression are patient-specific.

Towards a preclinical model

To facilitate future translational work applying DBS to this region, we targeted the A13 region using AAV8-CamKII-mCherry viruses. The CaMKIIα promoter virus is beneficial because it is biased towards excitatory cells (Haery et al., 2019), narrowing the diversity of transfected A13 region neurons and in our hands, the viral spread was contained within the A13 region. Optogenetic strategies have been used to activate retinal cells in humans, partially restoring visual function and providing optimism that AAV-based viral strategies can be adapted in other human brain regions (Sahel et al., 2021). A more likely possibility for stimulation of deep nuclei is that DREADD technology could be adapted, which would not require any implants, but this remains a longer-term possibility. In the short-term, our work suggests that the A13 is a possible target for

DBS. Gait dysfunction in PD is particularly difficult to treat, and indeed when DBS of subthalamic nucleus is deployed, a mixture of unilateral and bilateral approaches have been used (Lizarraga et al., 2016), along with stimulation of multiple targets (Stefani et al., 2007). This represents the heterogeneity of PD and underlines the need for considering multiple targets. In this regard, the identification of non-canonical dopaminergic pathways for the direct control of locomotion is promising (**Figure 6**). Our work highlights the A13 as a possible target, likely used in context and in concert with the activation of other identified targets.

Limitations

Currently, few PD animal models are available that adequately model the progression and the extent of SNc cellular degeneration while meeting the face validity of motor deficits (Dauer and Przedborski, 2003; Konnova et al., 2018). While the 6-OHDA models fail to capture the age-dependent chronic degeneration observed in PD, it provides additional advantages in providing robust motor deficits with acute degeneration and identifying compensatory changes compared to the unlesioned side. Moreover, it resembles the unilateral onset (Hughes et al., 1992) and persistent asymmetry (Lee et al., 1995) of motor dysfunction in PD. Another option could be the MPTP mouse model, which offers the ease of systemic administration and translational value to primate models; however, the motor deficits are variable and lack the asymmetry observed in human patients (Hughes et al., 1992; Jagmag et al., 2015; Lee et al., 1995; Meredith and Rademacher, 2011). Despite these limitations, the neurotoxin-based mouse models, such as MPTP and 6-OHDA, offer greater SNc cell loss than genetic-based models; in the case of the 6-OHDA model, it captures many aspects of motor dysfunctions in PD (Dauer and Przedborski, 2003; Jagmag et al., 2015; Konnova et al., 2018; Simola et al., 2007).

Conclusions

Parkinson's disease involves areas outside the classic nigrostriatal axis. Our work demonstrates that the A13 region drives locomotor activity and rescues bradykinetic and akinetic deficits caused by dysfunctional DAergic transmission in the basal ganglia. We show that A13 region-evoked locomotion has therapeutic potential for improving gait in advanced PD. Widespread remodelling of the A13 region connectome is critical to our understanding of the effects of dopamine loss in PD models. In summary, our findings support an exciting role for the A13 region in locomotion with demonstrated benefits in a mouse PD model and contribute to our understanding of heterogeneity in PD.

Materials and methods

Animals

All care and experimental procedures were approved by the University of Calgary Health Sciences Animal Care Committee (Protocol #AC19-0035). C57BL/6 male mice 49 - 56 days old (weight: M = 31.7 g, SEM = 2.0 g) were group-housed (≤ four per cage) on a 12-h light/dark cycle (07:00 lights on - 19:00 lights off) with ad libitum access to food and water, as well as cat's milk (Whiskas, Mars Canada Inc., Bolton, ON, Canada). Mice were randomly assigned to the groups described.

Surgical Procedures

We established a well-validated unilateral 6-OHDA mediated Parkinsonian mouse model (Thiele et al., 2012) (**Figure 1**, Movie S4). 30 minutes before stereotaxic microinjections, mice were intraperitoneally injected with desipramine hydrochloride (2.5 mg/ml, Sigma-Aldrich) and pargyline hydrochloride (0.5 mg/ml, Sigma-Aldrich) at 10 ml/kg (0.9% sterile saline, pH 7.4) to

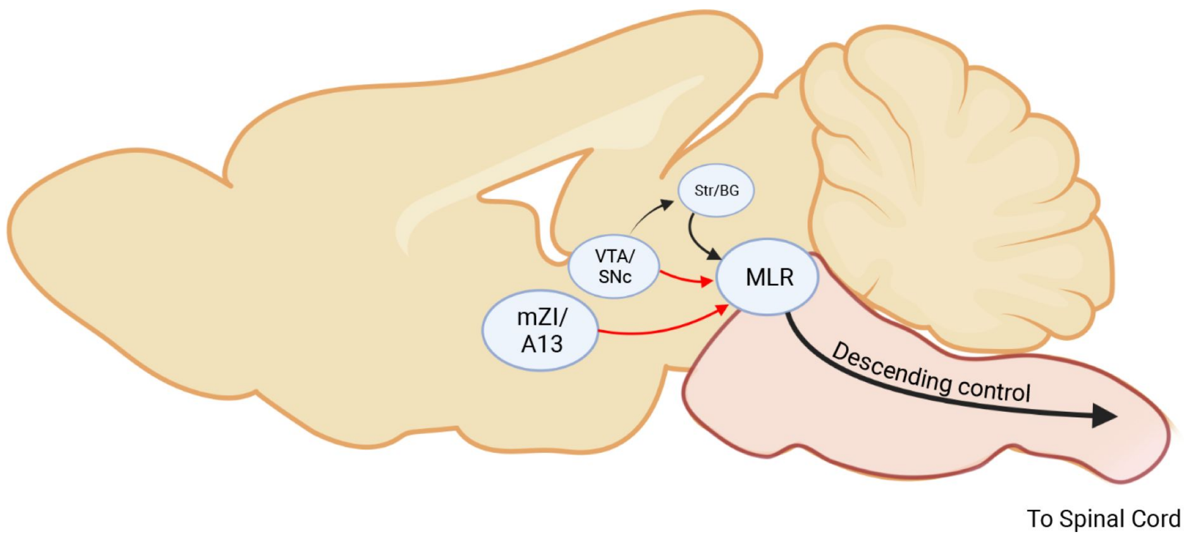

Figure 6

Comparing descending dopamine pathways for locomotor control.

Simplified connectivity map for the 3 dopamine pathways. The first pathway is the classical VTA/SNc projection to the striatum, and the SNr/GPi projects to the MLR. The VTA/SNc also directly projects to the MLR (Ryczko et al., 2016). The mZI/A13 region projects dopaminergic projections to the MLR (Sharma et al., 2018). Canonical pathways are in black, while non-canonical pathways are in red.

enhance selectivity and efficacy of 6-OHDA induced lesions (Thiele et al., 2012). All surgical procedures were performed using aseptic techniques, and mice were anesthetized using isoflurane (1 - 2%) delivered by 0.4 L/min of medical-grade oxygen (Vitalair 1072, 100% oxygen).

Mice were stabilized on a stereotaxic apparatus. Small craniotomies were made above the medial forebrain bundle (MFB) and the A13 nucleus within one randomly assigned hemisphere. Stereotaxic microinjections were performed using a glass capillary (Drummond Scientific, PA, USA; Puller Narishige, diameter 15 – 20 mm) and a Nanoject II apparatus (Drummond Scientific, PA, USA). 240 nL of 6-OHDA (3.6 µg, 15.0 mg/mL; Tocris, USA) was microinjected into the MFB (AP −1.2 mm from bregma; ML ±1.1 mm; DV −5.0 mm from the dura). Sham mice received a vehicle solution (240 nL of 0.2% ascorbic acid in 0.9% saline; Tocris, USA).

Whole Brain Experiments

For tracing purposes, a 50:50 mix of AAV8-CamKII-mCherry (Neurophotonics, Laval University, Quebec City, Canada, Lot #820, titre $2×10^{13}$ GC/ml) and AAVrg-CAG-GFP (Addgene, Watertown, MA, Catalogue #37825, Lot #V9234, titre ≥ $7×10^{12}$ vg/mL) was injected ipsilateral to 6-OHDA injections at the A13 nucleus in all mice (AP −1.22 mm from bregma; ML ±0.4 mm; DV −4.5 mm from the dura, the total volume of 110 nL at a rate of 23 nl/sec). Post-surgery care was the same for both sham and 6-OHDA injected mice. The animals were sacrificed 29 days after surgery.

Photoactivation Experiments

36.8 nL of AAVDJ-CaMKIIα-hChR2(H134R)-eYFP (UNC Stanford Viral Gene Core; Stanford, CA, US, Catalogue #AAV36; Lots #3081 and #6878, titres $1.9×10^{13}$ and $1.7×10^{13}$ GC/mL, respectively) or eYFP control virus (AAVDJ-CaMKIIα-eYFP; Lots #2958 & #5510, titres $7.64×10^{13}$ and $2.88×10^{13}$ GC/mL, respectively) was injected into the A13's stereotaxic coordinates (Sharma et al., 2018). A mono-fibre cannula (Doric Lenses, Quebec, Canada, Catalogue #B280-2401-5, MFC_200/230-0.48_5mm_MF2.5_FLT) was implanted slowly 300 µm above the viral injection site. Metabond® Quick Adhesive Cement System (C&B, Parkell, Brentwood, NY, US) and Dentsply Repair Material (Dentsply International Inc., York, PA, USA) were used to fix the ferrule in place. Animals recovered from the viral surgery for 19 days before follow-up behavioral testing. **Figure 1** illustrates the timeline of the behavioral tests.

ChR2 photoactivation

A Laserglow Technologies 473 nm laser and driver (LRS-0473-GFM-00100-05, North York, ON, Canada) were used to generate the photoactivation for experiments. The laser was triggered with TTL pulses from either an A.M.P.I. Master-8 stimulator (Jerusalem, Israel) or an Open Ephys PulsePal (Sanworks, Rochester, NY, US) set to 20 Hz with 10-ms pulse width. All fibre optic implants were tested for laser power before implantation (Thorlabs, Saint-Laurent, QC, Canada; optical power sensor (S130C) and meter (PM100D)). The Stanford Optogenetics irradiance calculator was used to estimate the laser power for stimulation ("Stanford Optogenetics Resource Center," n.d.). A 1×2 fibre-optic rotary joint (Doric Lenses, Quebec, Canada; FRJ_1x2i_FC-2FC_0.22) was used. The animals' behaviors were recorded with an overhead camera (SuperCircuits, Austin, TX, US; FRJ_1x2i_FC-2FC_0.22; 720 x 480 resolution; 30 fps). The video was processed online (Cleversys, Reston, VA; TopScan V3.0) with a TTL signal output from a National Instruments 24-line digital I/O box (NI, Austin, TX, US; USB-6501) to the Master-8 stimulator.

Behavioral Testing

Open Field Test

Each mouse was placed in a square arena measuring 70 (W) x 70 (L) x 50 (H) cm with opaque walls and recorded for 30 minutes using a vertically mounted video camera (Model PC165DNR, Supercircuits, Austin, TX, USA; 30 fps). 19 days following surgery, mice were habituated to the open field test (OFT) arena with a patch cable attached for three days in 30 minute sessions to bring animals to a common baseline of activity. On experimental days, after animals were placed in the OFT, a one-minute-on-three-minutes-off paradigm was repeated five times following an initial ten minutes baseline activity. Locomotion was registered when mice travelled a minimum distance of 10 cm at 6 cm/s for 20 frames over a 30-frame segment. When the mouse velocity dropped below 6 cm/s for 20 frames, locomotion was recorded as ending. Bouts of locomotion relate to the number of episodes where the animal met these criteria. Velocity data were obtained from the frame-by-frame results and further processed in a custom Python script to detect instantaneous speeds greater than 2 cm/s (Masini and Kiehn, 2022). All animals that had validated targeting of the A13 region were included in the OFT data presented in the results section, except for one sham ChR2 animal, which showed grooming rather than the typical locomotor phenotypes.

Pole Test

Mice were placed on a vertical wooden pole (50 cm tall and 1 cm diameter) facing upwards and then allowed to descend the pole into their home cage (Glajch et al., 2012). Animals were trained for three days and tested 2-5 days pre-surgery. Animals were acclimatized 21-22 days post-surgery under two conditions: without a patch cable and with the patch cable attached without photoactivation. On days 24-27, experimental trials were recorded with photoactivation. Video data were recorded for a minimum of three trials (Canon, Brampton, ON, Canada; Vixia HF R52; 1920 x 1080 resolution; 60 fps). A blinded scorer recorded the times for the following events: the hand release of the animal's tail, the animal fully turning to descend the pole, and the animal reaching the base of the apparatus. Additionally, partial falls, where the animal slipped down the pole but did not reach the base, and full falls, where the animal fell to the base, were recorded separately. All validated animals were included in the quantified data, including the sham ChR2 animal that began grooming in the OFT upon photoactivation. This animal displayed proficiency in performing the PT during photoactivation. It started grooming upon completion of the task when photoactivation was on. One sham ChR2 animal was photostimulated at 1 mW since it would jump off the apparatus at higher stimulation intensities.

Immunohistochemistry

A13 and SNc region

Post hoc analysis of the tissue was performed to confirm the 6-OHDA lesion and validate the targeting of the A13 region. Following behavioral testing, animals underwent a photoactivation protocol to activate neurons below the fibre optic tip (Koblinger et al., 2018). Animals were placed in an OFT for ten minutes before receiving three minutes of photoactivation. Ten minutes later, the animals were returned to their home cage. 90 minutes post photoactivation, animals were deeply anaesthetised with isoflurane and then transcardially perfused with room temperature PBS followed by cold 4% paraformaldehyde (PFA) (Sigma-Aldrich, Catalogue #441244-1KG). The animals were decapitated, and the whole heads were incubated overnight in 4% PFA at 4°C before the fibre optic was removed and the brain removed from the skull. The brain tissue was post-fixed for another 6 - 12 hours in 4% PFA at 4°C then transferred to 30% sucrose solution for 48 - 72 hours. The tissue was embedded in VWR® Clear Frozen Section Compound (VWR International LLC, Radnor, PA, US) and sectioned coronally at 40 or 50 µm using a Leica cryostat set to −21°C (CM 1850 UV, Concord, ON, Canada). Sections from the A13 region (−0.2 to −2.0 mm past bregma) and the SNc (−2.2 to −4.0 past bregma) were collected and stored in PBS containing 0.02% (w/v) sodium azide (EM Science, Catalogue #SX0299-1, Cherry Hill, NJ, US) (Keith B. J. Franklin and Paxinos, 2008).

Immunohistochemistry staining was done on free-floating sections. The A13 sections were labelled for c-Fos, TH, and GFP (to enhance eYFP viral signal), and received a DAPI stain to identify nuclei. The SNc sections were stained with TH and DAPI (**Table 1**). Sections were washed in PBS (3 x 10 mins) then incubated in a blocking solution comprised of PBS containing 0.5% Triton X-100 (Sigma-Aldrich, Catalogue #X100-500ML, St. Louis, MO, US) and 5% donkey serum (EMD Millipore, Catalogue #S30-100ML, Billerica, MA, USA) for 1 hour. This was followed by overnight (for SNc sections) or 24-hour (for A13 sections) incubation in a 5% donkey serum PBS primary solution at room temperature. On day 2, the tissue was washed in PBS (3 x 10 mins) before being incubated in a PBS secondary solution containing 5% donkey serum for 2 hours (for SNc tissue) or 4 hours (for A13 tissue). The secondary was washed with a PBS solution containing 1:1000 DAPI for 10 mins, followed by a final set of PBS washes (3 x 10 mins). Tissue was mounted on Superfrost® micro slides (VWR, slides, Radnor, PA, US) with mounting media (Vectashield®, Vector Laboratories Inc., Burlingame, CA, US), covered with #1 coverslips (VWR, Radnor, PA, US) then sealed.

Whole Brain

Mice were deeply anesthetized with isoflurane and transcardially perfused with PBS, followed by 4% PFA. To prepare for whole brain imaging, brains were first extracted and postfixed overnight in 4% PFA at 4°C. The next day, a modified iDISCO method (Renier et al., 2014) was used to clear the samples and perform quadruple immunohistochemistry in whole brains. The modifications include prolonged incubation and the addition of SDS for optimal labelling. The antibodies used are listed in **Table 1** and the protocol is provided in **Table 2**.

Image Acquisition and Analysis

Photoactivation Experiments

All tissue was initially scanned with an Olympus VS120-L100 Virtual Slide Microscope (UPlanSApo, 10x and 20x, NA = 0.4 and 0.75). Standard excitation and emission filter cube sets were used (DAPI, FITC, TRITC, Cy5), and images were acquired using an Orca Flash 4.0 sCMOS monochrome camera (Hamamatsu, Bridgewater Township, NJ, US). For c-Fos immunofluorescence, A13 sections of the tissue were imaged with a Leica SP8 FALCON (FAst Lifetime CONtrast) scanning confocal microscope equipped with a tunable laser and using a 63x objective (HC PlanApo, NA = 1.40).

SNc images were imported into Adobe Illustrator, where the SNc (Fougère et al., 2021), including the pars lateralis (SNl), was delineated using the TH immunostaining together with the medial lemniscus and cerebral peduncle as landmarks (bregma −3.09 and −3.68) (Iancu et al., 2005; Keith B. J. Franklin and Paxinos, 2008; Stott and Barker, 2014). Cell counts were obtained using a semi-automated approach using an Ilastik (v1.4.0b15) (Berg et al., 2019) trained model followed by corrections by a blinded counter (Fougère et al., 2021; Iancu et al., 2005). Targeting was confirmed on the 10x overview scans of the A13 region tissue by the presence of eYFP localized in the mZI around the A13 TH^+ nucleus, the fibre optic tip being visible near the mZI and A13 nucleus, and the presence of c-Fos positive cells in $ChR2^+$ tissue. C-Fos expression colocalization within the A13 region was performed using confocal images. The mZI & A13 region was identified with the 3rd ventricle and TH expression as markers (Keith B. J. Franklin and Paxinos, 2008).

Primary Antibody	Species	Type	Supplier	Catalogue #	Dilution	Tissue Region
c-Fos	Rabbit	Polyclonal	Synaptic Systems (Göttingen, Germany)	226 003	1:1000	A13
GFP	Chicken	Polyclonal	Aves Lab (Tigard, OR)	GFP-1010	1:1000 1:5000	Whole brain A13
mCherry	Rat	Monoclonal	Invitrogen, Thermo Fisher Scientific (Waltham, MA)	M11217	1:500	Whole brain
TH	Rabbit	Polyclonal	Abcam (Cambridge, UK)	AB112	1:500 1:1000	Whole brain SNc
TH	Sheep	Polyclonal	Millipore Sigma (Burlington, MA)	AB1542	1:500	A13
Secondary Antibody			**Supplier**	**Catalogue #**	**Dilution**	**Tissue Region**
Alexa Fluor® 488 Donkey Anti-Chicken			Jackson ImmunoResearch Inc. (Westgrove, PA)	703-545-155	1:200 1:1000	Whole brain A13
Alexa Fluor® 594 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-21207	1:500	A13
Alexa Fluor® 647 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-31573	1:1000	SNc
Alexa Fluor® 647 Donkey Anti-Sheep			Invitrogen, Thermo Fisher Scientific	A-21448	1:1000	A13
Alexa Fluor® 790 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-11374	1:200	Whole brain
Cy™3 AffiniPure Donkey Anti-Rat			Jackson ImmunoResearch Inc.	712-165-153	1:200	Whole brain
Nuclear Stain			**Supplier**	**Catalogue #**	**Dilution**	**Tissue Region**
DAPI			Invitrogen, Thermo Fisher Scientific	D1306	1:1000	A13, SNc
TO-PRO™-3 Iodide (642/661)			Invitrogen, Thermo Fisher Scientific	T3605	1:5000	Whole brain

Table 1.

List of antibodies used for immunohistochemical staining of the A13 and SNc regions, as well as the whole brain.

Day #	Instructions
1	Washed 2 x 30 minutes in PBS on a shaker at room temperature.
2	Dehydrated tissue in methanol/H_2O series of 20%, 40%, 60%, 80%, 100%, 100% (1 hour each at room temperature) then left overnight in a 66% dichloromethane/33% methanol solution.
3	Washed twice in 100% methanol at room temperature and then chilled the samples at 4°C. Bleached the samples in chilled fresh 5% H_2O_2 in methanol (1 volume 30% H_2O_2 to 5 volumes methanol), overnight at 4°C.
4	Rehydrated in methanol/water series of 80%, 60%, 40%, 20%, PBS (1 hour each at room temperature). Washed twice in PTx.2 (0.2% Triton™ X-100 in PBS). Incubated the samples in Permeabilization Solution at 37°C for 2 days.
6	Washed 3 x 1-2 hours in 0.5mM SDS/PBS at 37°C then incubated for 3 days.
9	Incubated with the primary antibody in 0.5mM SDS/PBS at 37°C for 3 days.
12	Refreshed with the primary antibody in PTx.2 at 37°C then incubated for 4 days.
17	Washed 5 x 2 hours in PTwH (0.2% Tween® 20 and 10 mg/mL Heparin stock solution in PBS) and incubated at 37°C overnight.
18	Incubated with secondary antibody in PTwH/3% Donkey Serum at 37°C for 3 days.
21	Refreshed with secondary antibody in PTwH/3% Donkey Serum at 37°C for 4 days.
25	Washed 5 x 2 hours in PTwH then incubated at 37°C overnight.
26	Dehydrated in methanol/water series of 20%, 40%, 60%, 80%, 100% (1 hour each at room temperature). Refreshed the samples with 100 % methanol and left overnight at room temperature.
27	Incubated in 66% DCM/33% methanol for 3 hours on a shaker at room temperature. Then incubated in 100% DCM (Sigma 270997-12X100ML) for 2 x 15 minutes (with shaking) to wash away methanol. Incubated samples with ethyl cinnamate for 3 hours at room temperature with shaking. Refreshed ethyl cinnamate then left at room temperature for imaging.

Table 2.

Protocol for Whole Brain Clearing.

Whole Brain Experiments

Cleared whole brain samples were imaged using a light-sheet microscope (LaVision Biotech UltraMicroscope, LaVision, Bielefeld, Germany) with an Olympus MVPLAPO 2x objective with 4x optical zoom (NA = 0.475) and a 5.7 mm dipping cap that is adjusted for the high refractive index of 1.56. The brain samples were imaged in an ethyl cinnamate medium to match the refractive indices and illuminated by three sheets of light bilaterally. Each light sheet was 5 μm thick, and the width was set at 30% to ensure sufficient illumination at the centroid of the sample. Laser power intensities and chromatic aberration corrections used for each laser were as follows: 10% power for 488 nm laser, 5% power for 561 nm laser with 780 nm correction, 40% power for 640 nm laser with 960 nm correction, and 100% power for 785 nm laser with 1,620 nm correction. Each sample was imaged coronally in 8 by 6 squares with 20% overlap (10,202 μm by 5,492 μm in total) and a z-step size of 15 μm (xyz resolution = 0.813 μm x 0.813 μm x 15 μm). While an excellent choice for our work, confocal microscopy offers better resolution at the expense of time. To gain a better resolution using a light-sheet microscope in select regions (eg. SNc and A13 cells), we increased the optical zoom to 6.3x.

A13 Connectome Analysis

Images were processed using ImageJ software (Schneider et al., 2012). Raw images were stitched, and a z-encoded maximum intensity projection across a 90 μm thick optical section was obtained across each brain. 90 μm sections were chosen because the 2008 Allen reference atlas images are spaced out at around 100 μm. Brains with insufficient quality in labeling were excluded from analysis (n = 1 of three sham and n = 3 of six 6-OHDA mice). Instructions for identifying YFP^+ or TH^+ cells to annotate were provided to the manual counters. YFP^+ and TH^+ cells were manually counted using the Cell Counter Plug-In (ImageJ). $mCherry^+$ fibers were segmented semi-automatically using Ilastik software (Berg et al., 2019) and quantified using particle analysis in ImageJ. Images and segmentations were imported into WholeBrain software to be registered with the 2008 Allen reference atlas (Fürth et al., 2018). The TO-PRO™-3 and TH channels were used as reference channels to register each section to a corresponding atlas image. ImageJ quantifications of cell and fiber segmentations were exported in XML formats and registered using WholeBrain software. To minimize the influence of experimental variation on the total labeling of neurons and fibers, the afferent cell counts or efferent fiber areas in each brain region were column divided by the total number found in a brain to obtain the proportion of total inputs and outputs. Connectome analyses were performed using custom R scripts (L. H. Kim et al., 2021). For interregional correlation analyses, the data were normalized to a $\log_{10}$ value to reduce variability and bring brain regions with high and low proportions of cells and fibers to a similar scale. The consistency of afferent and efferent proportions between mice was compared in a pairwise manner using Spearman's correlation (**Figure S5**).

Quantification of 6-OHDA mediated TH^+ cell loss

The percentage of TH^+ cell loss was quantified to confirm 6-OHDA mediated SNc lesions. TH^+ cells within ZI, VTA and SNc areas from 90 μm thick optical brain slice images (AP: −0.655 to −3.88 mm from bregma) were manually counted by two blinded counters (n = 3 sham and n = 6 6-OHDA mice; ZI region in 2 of 6 6-OHDA mice were excluded due to presence of abnormal scarring/healing at the injection site of viruses). Subsequently, WholeBrain software (Fürth et al., 2018) was used to register and tabulate TH^+ cells in the contralesional and ipsilesional brain regions of interest. Counts obtained from the two counters were averaged per region. The percentage of TH^+ cell loss was calculated by dividing the difference in counts between contralesional and ipsilesional sides by the contralesional side count and multiplying by 100%.

Statistical analyses

All data were tested for normality using a Shapiro-Wilk test to determine the most appropriate statistical tests. The percent ipsilesional TH^+ neuron loss within the SNc as defined above using a Pearson correlation (Fougère et al., 2021) was used to ascertain the effect of the 6-OHDA lesion on behavior. A Wilcoxon rank-sum test was performed for comparisons within subjects at two timepoints where normality failed, and the central limit theorem could not be applied. The two groups were compared using an unpaired t-test with Welch's correction. A mixed model ANOVA (MM ANOVA) was used to compare the effects of group type, injection type and time. Additionally, Mauchly's test of sphericity was performed to account for differences in variability within the repeated measures design. A Greenhouse Geisser correction was applied to all ANOVAs where Mauchly's test was significant for RM and MM ANOVAs. The *post hoc* multiple comparisons were run when the respective ANOVAs reached significance using Dunnett's or Dunn's tests for repeated measures of parametric and non-parametric tests, respectively. The pre-stimulation timepoints were used as the control time point to determine if stimulation altered behavior. A Bonferroni correction was added for *post hoc* comparisons following a MM ANOVA between groups at given time points to control for alpha value inflation. All correlations, t-tests, and ANOVAs were performed, and graphs were created using Prism version 9.3.1 (Graphpad) or SPSS (IBM, 28.0.1.0). Full statistical reporting is in Supplemental Statistics.xls.

Figures

Figures were constructed using Adobe Photoshop, Illustrator, and Biorender.

Author Contributions

LHK and AL performed experiments and prepared figures. MAT and PJW edited figures. LHK, AL, ZHT and PJW conceived and designed the research and interpreted the results. PJW procured funding for the experiments. SS and AL performed surgeries for lesions, optogenetic experiments and conducted behavioral experiments. LHK and SEAE optimized light-sheet imaging. MAT, ST, and CM performed manual cell counting. TC performed analysis and prepared figures on gait analysis. LHK, AL, TC, ZHT, and PJW drafted the manuscript. All authors reviewed and approved the final version of the manuscript.

Competing Interest Statement

None.

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Data availability

All datasets and code will be made available on a public repository.

Supplementary material

Movie S1. Photoactivation of the A13 region in a 6-OHDA model mouse producing increased locomotion in the OFT (2x speed).

Movie S2. Photoactivation of the A13 region in a sham mouse producing increased locomotion in the OFT (2x speed).

Movie S3. Photoactivation of the A13 region during the pole test in a 6-OHDA model mouse decreases pole descent time (0.5x speed).

Movie S4. Video showing TH staining following whole brain imaging and staining in a 6-OHDA model mouse brain. Focus on TH expression in the A13 and SNc regions.

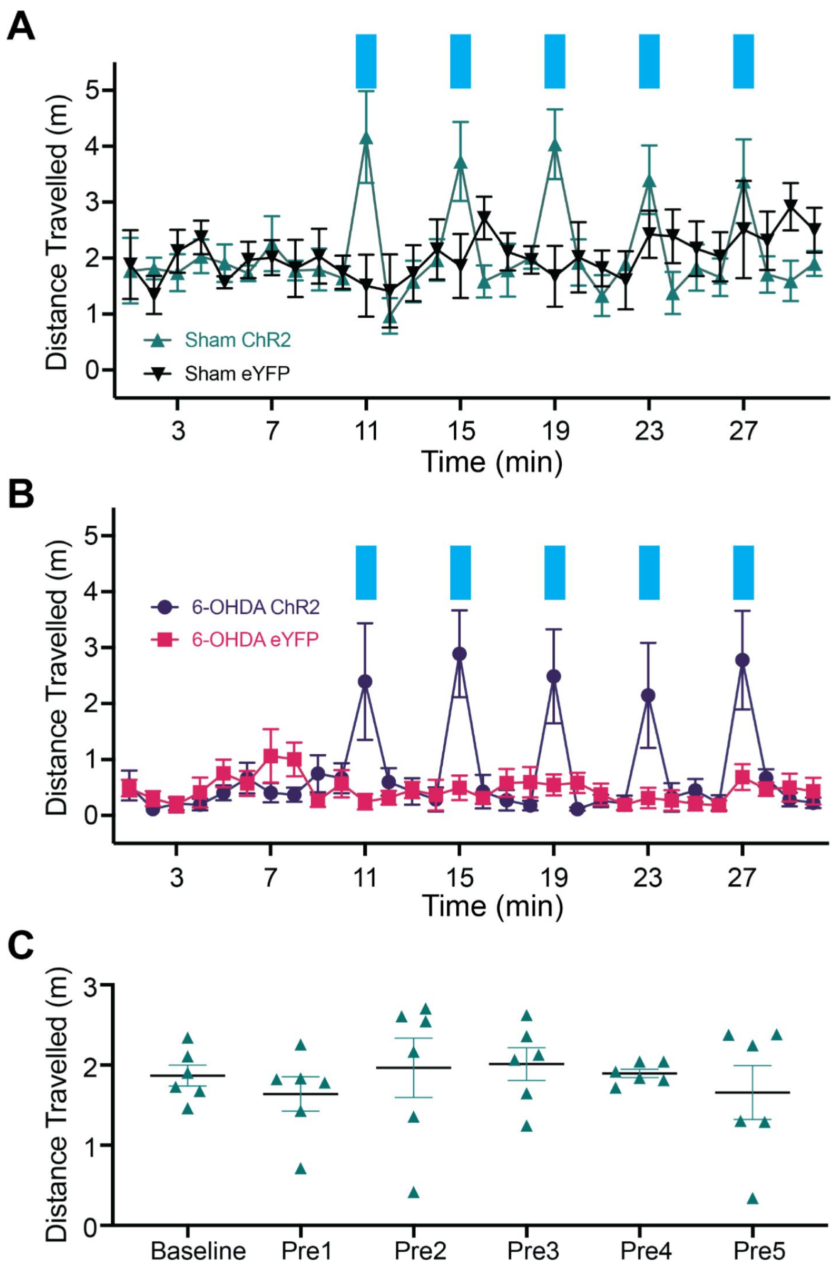

Figure S1.

Time course of open field locomotion distance travelled over 30 minutes.

(A-B) 30-minute open field experiment group averages for sham **(A)** and 6-OHDA **(B)** animals with photoactivation plotted as 1-minute bins of distance travelled. Blue bars indicate 1-minute trials with photoactivation. (**C)** Locomotion distance travelled for the six sham ChR2 animals at baseline and at the five pre-timepoints compared using a 1-way RM ANOVA ($F_{5,25}$ = 0.486, P = .783). Data indicate mean ± SEM bars.

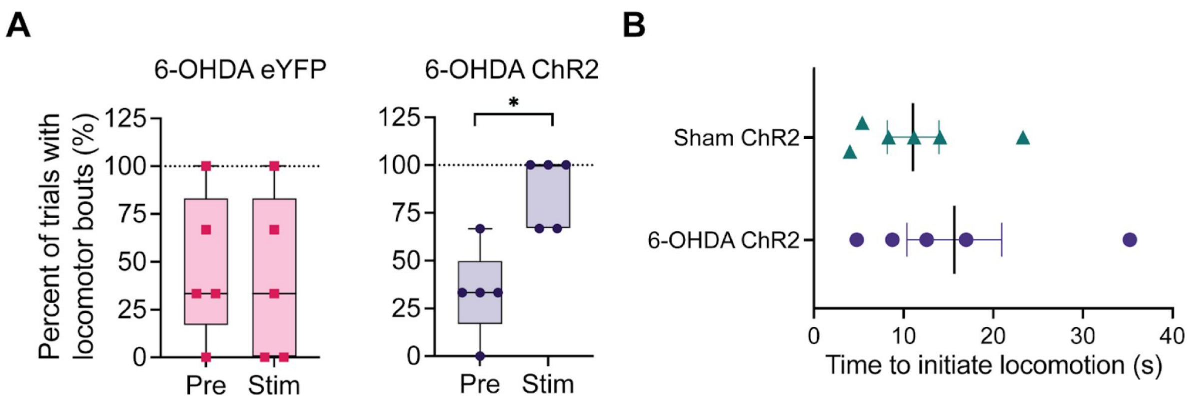

Figure S2.

Characterization of A13 region photoactivation temporal dynamics on locomotion initiation. (A)

Percent of trials where there was at least one bout of locomotion. Data are plotted as box and whiskers with the horizontal line through the box indicating the group median, interquartile range indicated by the limits of the box, and group minimum and maximum indicated by the whiskers. **(B)** The average time for the ChR2 group animals to begin locomotion after the onset of photoactivation. Means plotted with error bars indicating ± SEM. Asterisks indicate significant comparisons using the Wilcoxon signed-rank test: ** $P \leq 0.05$.

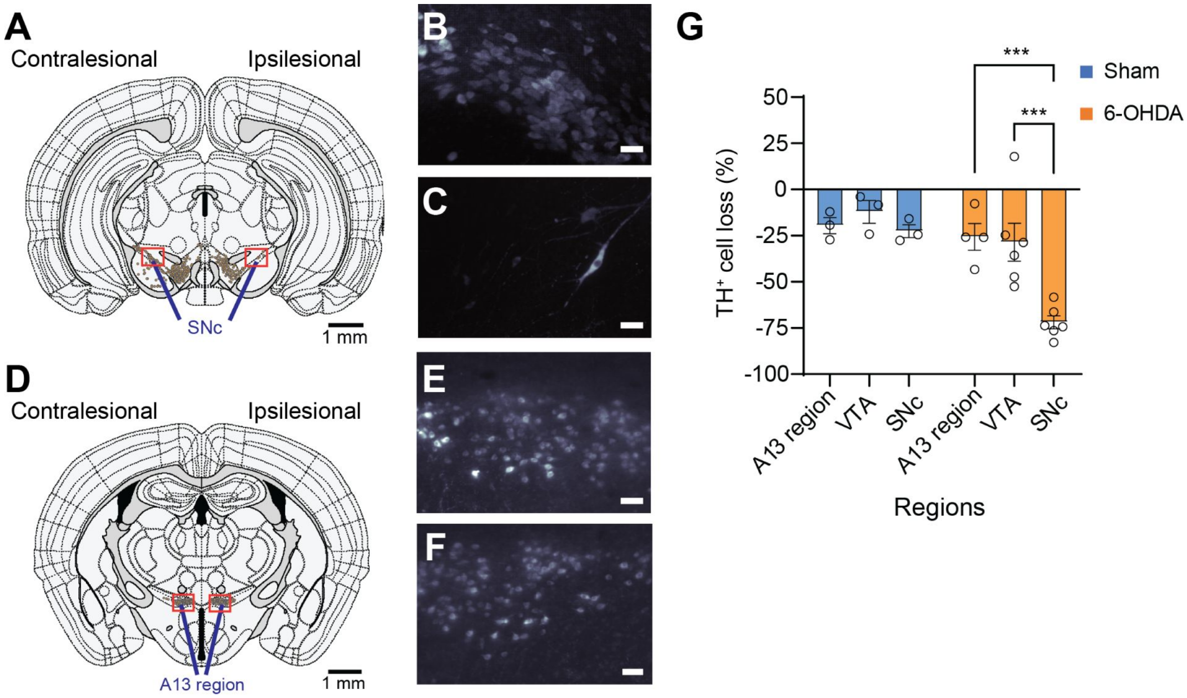

Figure S3.

Preservation of TH^+ A13 cells in Parkinsonian mouse models.

Representative slices of SNc (AP: −3.08 mm, **A**) and A13 region (AP: −1.355 mm, **D**) following registration with WholeBrain software 64. Full 3D brain is available (see Movie S4). There was a lack of TH^+ SNc cells following 6-OHDA injections at the MFB **(A)**. **(B, C)** Zoomed sections (90 μm thickness) of red boxes in panel A in left to right order. Meanwhile, TH^+ VTA cells were preserved bilaterally. In addition, TH^+ A13 cells were present ipsilesional to 6-OHDA injections **(D)**. **(E, F)** Zoomed sections (90 μm thickness) of red boxes in panel D in left to right order. Scale bars are 50 μm. When calculating the percentage of TH^+ cell loss normalized to the intact side, there was a significant interaction between the condition group and brain regions (repeated measures two-way ANOVA with *post hoc* Bonferroni pairwise, sham: n = 3, 6-OHDA: n = 6) **G**). 6-OHDA treated mice showed a significantly greater percentage of TH^+ cell loss in SNc compared to VTA and A13 region (VTA vs. SNc: P = 0.005; A13 region vs. SNc: P = 0.029). In contrast, sham showed no significant difference in TH^+ cell loss across SNc, VTA and A13 region regions (P > 0.05). *$P \leq 0.05$, and **$P \leq 0.01$. Scale bars are 50 μm unless otherwise indicated.

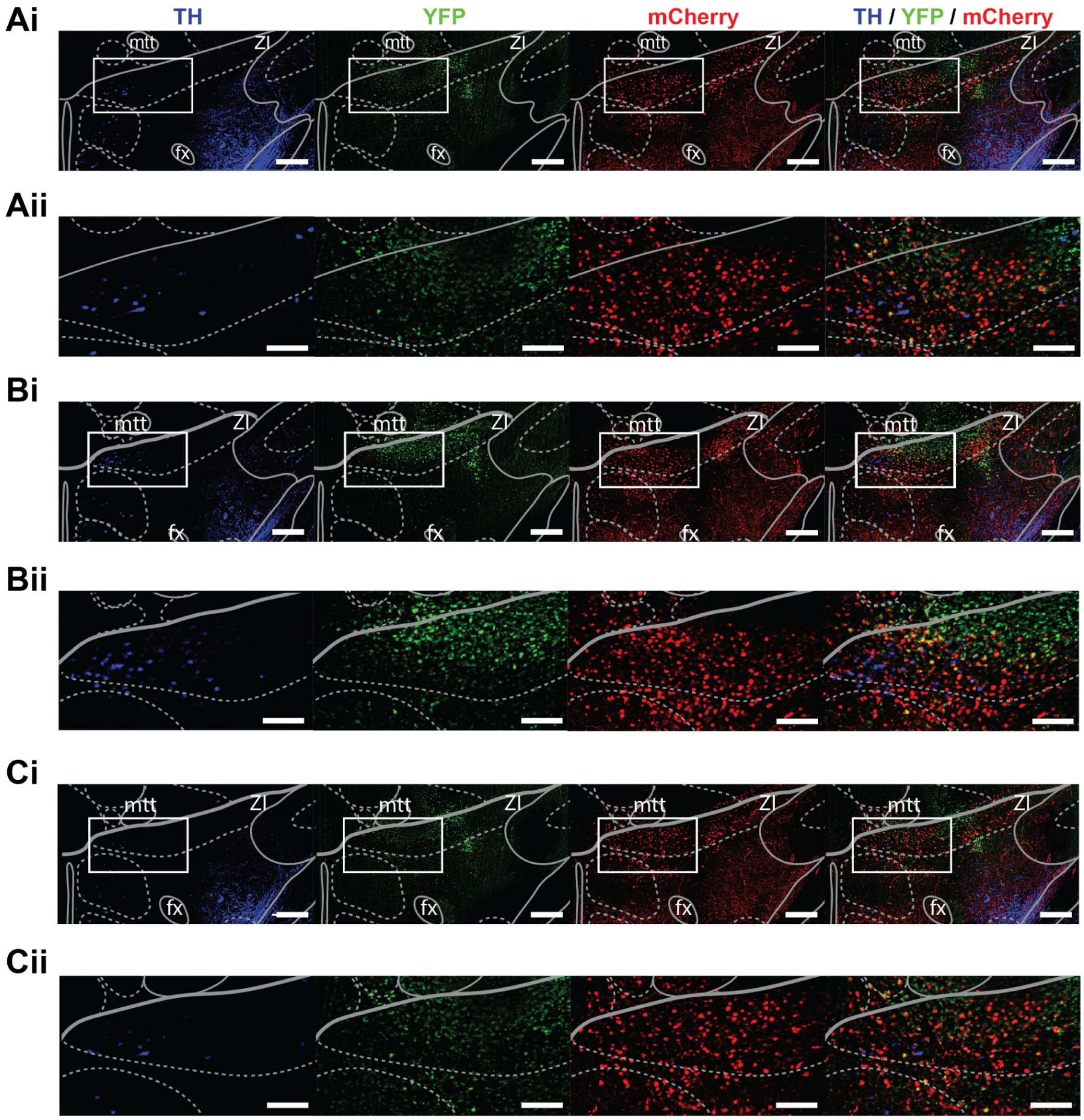

Figure S4.

Example of the injection core in a sham brain for viral tracers and the rostral and caudal spread to the injection site (A13). Viral tracers (AAV8-CamKII-mCherry and AAVrg-CAG-GFP) were mixed 50:50. Light-sheet images around the injection site were obtained with 2x objective, 6.3x optical zoom, and a z-step size of 2 µm (xyz resolution = 0.477 µm x 0.477 µm x 2 µm). Background filtering (median value of 20 pixels and Gaussian smoothing with a sigma value of 10) was performed in ImageJ software 1 and visualized in IMARIS 9.8 (Belfast, United Kingdom). 2008 Allen reference atlas images were overlaid on top of 90 µm maximum intensity projections taken from IMARIS 9.8 (Belfast, United Kingdom):-1.26 mm **(A)**, −1.36 mm **(B)**, and −1.46 mm **(C)**. Zoomed in sections of each white rectangular area at each coordinate (rows '**i**') are displayed below for each fluorophore (rows '**ii**'). Scale bars for rows 'i' are 200 µm and for rows 'ii are 100 µm.

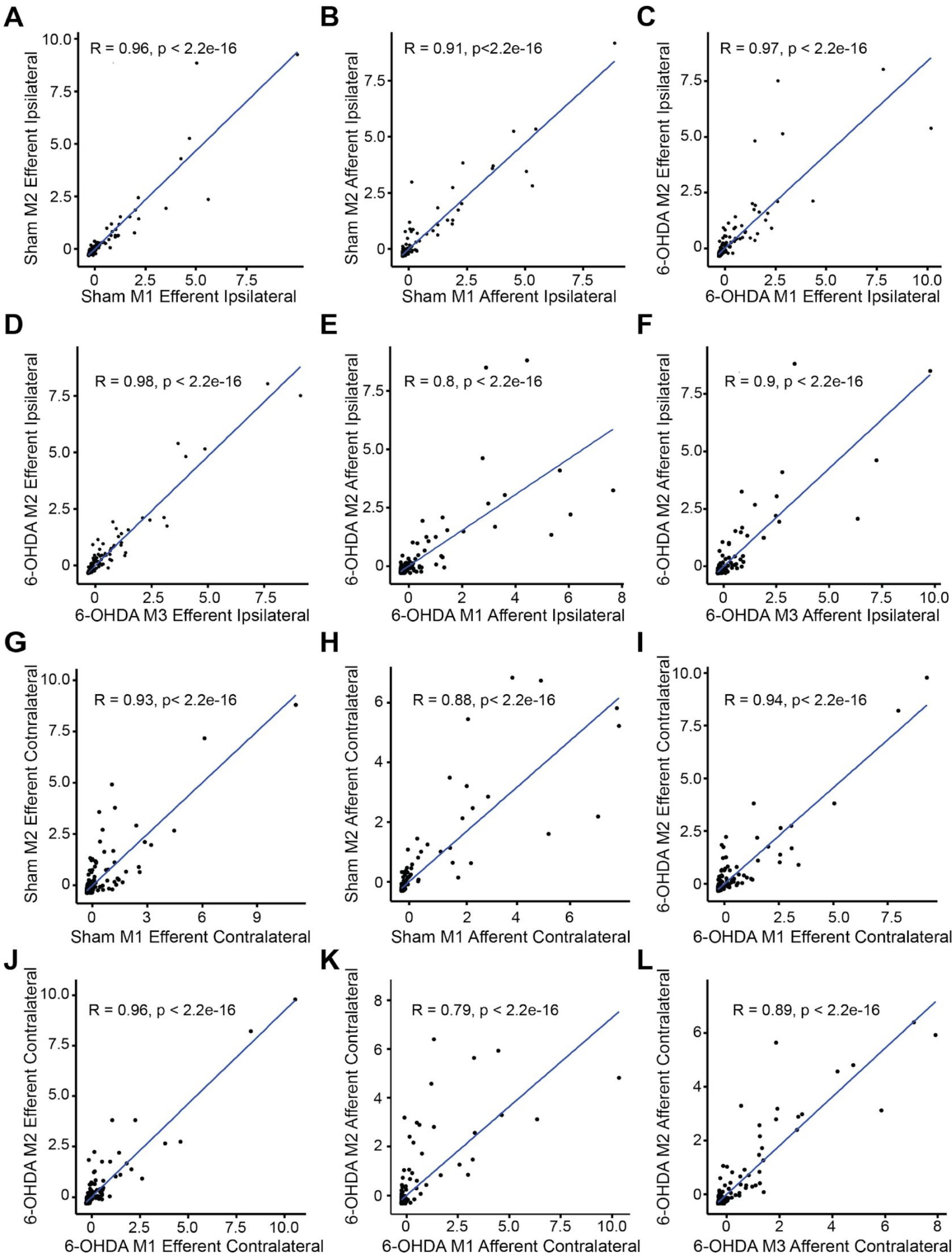

Figure S5.

The consistency of afferent and efferent proportions across mice was compared in a pairwise manner. An experimental variation on the total labeling of neurons and fibers was minimized by dividing the afferent cell counts or efferent fiber areas in each brain region by the total number found in a brain to obtain the proportion of total inputs and outputs. Using Spearman's correlation analysis, we found afferent and efferent proportions across animals to be consistent among each other with an average correlation of 0.91 (SEM = 0.02). M1 = mouse #1, M2 = mouse #2, M3 = mouse #3.

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Reviewer #1 (Public Review):

Summary:
This study aimed to investigate the effects of optically stimulating the A13 region in healthy mice and a unilateral 6-OHDA mouse model of Parkinson's disease (PD). The primary objectives were to assess changes in locomotion, motor behaviors, and the neural connectome. For this, the authors examined the dopaminergic loss induced by 6-OHDA lesioning. They found a significant loss of tyrosine hydroxylase (TH+) neurons in the substantia nigra pars compacta (SNc) while the dopaminergic cells in the A13 region were largely preserved. Then, they optically stimulated the A13 region using a viral vector to deliver the channelrhodopsine (CamKII promoter). In both sham and PD model mice, optogenetic stimulation of the A13 region induced pro-locomotor effects, including increased locomotion, more locomotion bouts, longer durations of locomotion, and higher movement speeds. Additionally, PD model mice exhibited increased ipsilesional turning during A13 region photoactivation. Lastly, the authors used whole-brain imaging to explore changes in the A13 region's connectome after 6-OHDA lesions. These alterations involved a complex rewiring of neural circuits, impacting both afferent and efferent projections. In summary, this study unveiled the pro-locomotor effects of A13 region photoactivation in both healthy and PD model mice. The study also indicates the preservation of A13 dopaminergic cells and the anatomical changes in neural circuitry following PD-like lesions that represent the anatomical substrate for a parallel motor pathway.

Strengths:
These findings hold significant relevance for the field of motor control, providing valuable insights into the organization of the motor system in mammals. Additionally, they offer potential avenues for addressing motor deficits in Parkinson's disease (PD). The study fills a crucial knowledge gap, underscoring its importance, and the results bolster its clinical relevance and overall strength.

The authors adeptly set the stage for their research by framing the central questions in the introduction, and they provide thoughtful interpretations of the data in the discussion section. The results section, while straightforward, effectively supports the study's primary conclusion - the pro-locomotor effects of A13 region stimulation, both in normal motor control and in the 6-OHDA model of brain damage.

Weaknesses:

1. Anatomical investigation. I have a major concern regarding the anatomical investigation of plastic changes in the A13 connectome (Figures 4 and 5). While the methodology employed to assess the connectome is technically advanced and powerful, the results lack mechanistic insight at the cell or circuit level into the pro-locomotor effects of A13 region stimulation in both physiological and pathological conditions. This concern is exacerbated by a textual description of results that doesn't pinpoint precise brain areas or subareas but instead references large brain portions like the cortical plate, making it challenging to discern the implications for A13 stimulation. Lastly, the study is generally well-written with a smooth and straightforward style, but the connectome section presents challenges in readability and comprehension. The presentation of results, particularly the correlation matrices and correlation strength, doesn't facilitate biological understanding. It would be beneficial to explore specific pathways responsible for driving the locomotor effects of A13 stimulation, including examining the strength of connections to well-known locomotor-associated regions like the Pedunculopontine nucleus, Cuneiformis nucleus, LPGi, and others in the diencephalon, midbrain, pons, and medulla. Additionally, identifying the primary inputs to A13 associated with motor function would enhance the study's clarity and relevance.

The study raises intriguing questions about compensatory mechanisms in Parkinson's disease and a new perspective on the preservation of dopaminergic cells in A13, despite the SNc degeneration, and the plastic changes to input/output matrices. To gain inspiration for a more straightforward reanalysis and discussion of the results, I recommend the authors refer to the paper titled "Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon from the David Kleinfeld laboratory." This could guide the authors in investigating motor pathways across different brain regions.

1. Description of locomotor performance. Figure 3 provides valuable data on the locomotor effects of A13 region photoactivation in both control and 6-OHDA mice. However, a more detailed analysis of the changes in locomotion during stimulation would enhance our understanding of the pro-locomotor effects, especially in the context of 6-OHDA lesions. For example, it would be informative to explore whether the probability of locomotion changes during stimulation in the control and 6-OHDA groups. Investigating reaction time, speed, total distance, and even kinematic aspects during stimulation could reveal how A13 is influencing locomotion, particularly after 6-OHDA lesions. The laboratory of Whelan has a deep knowledge of locomotion and the neural circuits driving it so these features may be instructive to infer insights on the neural circuits driving movement. On the same line, examining features like the frequency or power of stimulation related to walking patterns may help elucidate whether A13 is engaging with the Mesencephalic Locomotor Region (MLR) to drive the pro-locomotor effects. These insights would provide a more comprehensive understanding of the mechanisms underlying A13-mediated locomotor changes in both healthy and pathological conditions.

Reviewer #2 (Public Review):

Summary:
The paper by Kim et al. investigates the potential of stimulating the dopaminergic A13 region to promote locomotor restoration in a Parkinson's mouse model. Using wild-type mice, 6-OHDA injection depletes dopaminergic neurons in the substantia nigra pars compacta, without impairing those of the A13 region and the ventral tegmentum area, as previously reported in humans. Moreover, photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region improves bradykinesia and akinetic symptoms after 6-OHDA injection. Whole-brain imaging with retrograde and anterograde tracers reveals that the A13 region undergoes substantial changes in the distribution of its afferents and projections after 6-OHDA injection. The study suggests that if the remodeling of the A13 region connectome does not promote recovery following chronic dopaminergic depletion, photostimulation of the A13 region restores locomotor functions.

Strengths:
Photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region promotes locomotion and locomotor recovery of wild-type mice 1 month after 6-OHDA injection in the medial forebrain bundle, thus identifying a new potential target for restoring motor functions in Parkinson's disease patients.

Weaknesses:
Electrical stimulation of the medial Zona Incerta, in which the A13 region is located, has been previously reported to promote locomotion (Grossman et al., 1958). Recent mouse studies have shown that if optogenetic or chemogenetic stimulation of GABAergic neurons of the Zona Incerta promotes and restores locomotor functions after 6-OHDA injection (Chen et al., 2023), stimulation of glutamatergic ZI neurons worsens motor symptoms after 6-OHDA (Lie et al., 2022).

Although CAMKIIa is a marker of presumably excitatory neurons and can be used as an alternative marker of dopaminergic neurons, behavioral results of this study raise questions about the neuronal population targeted in the vicinity of the A13 region. Moreover, if YFP and CHR2-YFP neurons express dopamine (TH) within the A13 region (Fig. 2), there is also a large population of transduced neurons within and outside of the A13 region that do not, thus suggesting the recruitment of other neuronal cell types that could be GABAergic or glutamatergic.

Regarding the analysis of interregional connectivity of the A13 region, there is a lack of specificity (the viral approach did not specifically target the A13 region), the number of mice is low for such correlation analyses (2 sham and 3 6-OHDA mice), and there are no statistics comparing 6-OHDA versus sham (Fig. 4) or contra- versus ipsilesional sides (Fig. 5). Moreover, the data are too processed, and the color matrices (Fig. 4) are too packed in the current format to enable proper visualization of the data. The A13 afferents/efferents analysis is based on normalized relative values; absolute values should also be presented to support the claim about their upregulation or downregulation.

In the absence of changes in the number of dopaminergic A13 neurons after 6-OHDA injection, results from this correlation analysis are difficult to interpret as they might reflect changes from various impaired brain regions independently of the A13 region. There is no causal link between anatomical and behavioral data, which raises questions about the relevance of the anatomical data.

Overall, the study does not take advantage of genetic tools accessible in the mouse to address the direct or indirect behavioral and anatomical contributions of the A13 region to motor control and recovery after 6-OHDA injection.

Reviewer #3 (Public Review):

Kim, Lognon et al. present an important finding on pro-locomotor effects of optogenetic activation of the A13 region, which they identify as a dopamine-containing area of the medial zona incerta that undergoes profound remodeling in terms of afferent and efferent connectivity after administration of 6-OHDA to the MFB. The authors claim to address a model of PD-related gait dysfunction, a contentious problem that can be difficult to treat with dopaminergic medication or DBS in conventional targets. They make use of an impressive array of technologies to gain insight into the role of A13 remodeling in the 6-OHDA model of PD. The evidence provided is solid and the paper is well written, but there are several general issues that reduce the value of the paper in its current form, and a number of specific, more minor ones. Also, some suggestions, that may improve the paper compared to its recent form, come to mind.

The most fundamental issue that needs to be addressed is the relation of the structural to the behavioral findings. It would be very interesting to see whether the structural heterogeneity in afferent/effects projections induced by 6-OHDA is related to the degree of symptom severity and motor improvement during A13 stimulation.

The authors provide extensive interrogation of large-scale changes in the organization of the A13 region afferent and efferent distributions. It remains unclear how many animals were included to produce Fig 4 and 5. Fig S5 suggests that only 3 animals were used, is that correct? Please provide details about the heterogeneity between animals. Please provide a table detailing how many animals were used for which experiment. Were the same animals used for several experiments?

While the authors provide evidence that photoactivation of the A13 is sufficient in driving locomotion in the OFT, this pro-locomotor effect seems to be independent of 6-OHDA-induced pathophysiology. Only in the pole test do they find that there seems to be a difference between Sham vs 6-OHDA concerning the effects of photoactivation of the A13. Because of these behavioral findings, optogenic activation of A13 may represent a gain of function rather than disease-specific rescue. This needs to be highlighted more explicitly in the title, abstract, and conclusion.

The authors claim that A13 may be a possible target for DBS to treat gait dysfunction. However, the experimental evidence provided (in particular the lack of disease-specific changes in the OFT) seems insufficient to draw such conclusions. It needs to be highlighted that optogenetic activation does not necessarily have the same effects as DBS (see the recent review from Neumann et al. in Brain: https://pubmed.ncbi.nlm.nih.gov/37450573/). This is important because ZI-DBS so far had very mixed clinical effects. The authors should provide plausible reasons for these discrepancies. Is cell-specificity, which only optogenetic interventions can achieve, necessary? Can new forms of cyclic burst DBS achieve similar specificity (Spix et al, Science 2021)? Please comment.

In a recent study, Jeon et al (Topographic connectivity and cellular profiling reveal detailed input pathways and functionally distinct cell types in the subthalamic nucleus, 2022, Cell Reports) provided evidence on the topographically graded organization of STN afferents and McElvain et al. (Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon, 2021, Neuron) have shown similar topographical resolution for SNr efferents. Can a similar topographical organization of efferents and afferents be derived for the A13/ ZI in total?

In conclusion, this is an interesting study that can be improved by taking into consideration the points mentioned above.

Author Response

We appreciate the editor's and reviewers' time to review our manuscript. We will work on the suggestions and have provided an initial assessment of what we can do for our revised submission.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *This study aimed to investigate the effects of optically stimulating the A13 region in healthy mice and a unilateral 6-OHDA mouse model of Parkinson's disease (PD). The primary objectives were to assess changes in locomotion, motor behaviors, and the neural connectome. For this, the authors examined the dopaminergic loss induced by 6-OHDA lesioning. They found a significant loss of tyrosine hydroxylase (TH+) neurons in the substantia nigra pars compacta (SNc) while the dopaminergic cells in the A13 region were largely preserved. Then, they optically stimulated the A13 region using a viral vector to deliver the channelrhodopsine (CamKII promoter). In both sham and PD model mice, optogenetic stimulation of the A13 region induced pro-locomotor effects, including increased locomotion, more locomotion bouts, longer durations of locomotion, and higher movement speeds. Additionally, PD model mice exhibited increased ipsilesional turning during A13 region photoactivation. Lastly, the authors used whole-brain imaging to explore changes in the A13 region's connectome after 6-OHDA lesions. These alterations involved a complex rewiring of neural circuits, impacting both afferent and efferent projections. In summary, this study unveiled the pro-locomotor effects of A13 region photoactivation in both healthy and PD model mice. The study also indicates the preservation of A13 dopaminergic cells and the anatomical changes in neural circuitry following PD-like lesions that represent the anatomical substrate for a parallel motor pathway.*
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> *These findings hold significant relevance for the field of motor control, providing valuable insights into the organization of the motor system in mammals. Additionally, they offer potential avenues for addressing motor deficits in Parkinson's disease (PD). The study fills a crucial knowledge gap, underscoring its importance, and the results bolster its clinical relevance and overall strength.*
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> *The authors adeptly set the stage for their research by framing the central questions in the introduction, and they provide thoughtful interpretations of the data in the discussion section. The results section, while straightforward, effectively supports the study's primary conclusion the pro-locomotor effects of A13 region stimulation, both in normal motor control and in the 6-OHDA model of brain damage.*

We thank the reviewer for their positive comments.

> *Weaknesses:*

1. *Anatomical investigation. I have a major concern regarding the anatomical investigation of plastic changes in the A13 connectome (Figures 4 and 5). While the methodology employed to assess the connectome is technically advanced and powerful, the results lack mechanistic insight at the cell or circuit level into the pro-locomotor effects of A13 region stimulation in both physiological and pathological conditions. This concern is exacerbated by a textual description of results that doesn't pinpoint precise brain areas or subareas but instead references large brain portions like the cortical plate, making it challenging to discern the implications for A13 stimulation. Lastly, the study is generally well-written with a smooth and straightforward style, but the connectome section presents challenges in readability and comprehension. The presentation of results, particularly the correlation matrices and correlation strength, doesn't facilitate biological understanding. It would be beneficial to explore specific pathways responsible for driving the locomotor effects of A13 stimulation, including examining the strength of connections to well-known locomotor-associated regions like the Pedunculopontine nucleus, Cuneiformis nucleus, LPGi, and others in the diencephalon, midbrain, pons, and medulla.*

We considered two approaches initially. The first approach was to look at specific projections to the motor regions, focusing on the MLR. The second approach was to utilize a whole-brain analysis that is presented here. Given what we know about the zona incerta, especially its integrative role, we felt that a reasonable starting point was to examine the full connectome. The value of the whole-brain approach is that it provides a high-level overview of the afferents and efferents to the region. The changes in the brain that occur following Parkinson-like lesions, such as those in the nigrostriatal pathway, are known to be complex and can affect neighbouring regions such as the A13. Therefore, we wished to highlight the A13, which we considered a therapeutic target, and examine changes in connectivity that could occur following acute lesions affecting the SNc. We acknowledge that this study does not provide a causal link, but it presents the fundamental background information for subsequent hypothesis-driven, focused, region-specific analysis.

The terms provided were from the Allen Brain Atlas terminology and were presented as abbreviations. We have looked at other ways to present it, including a greater emphasis on raw numbers and highlighting motor-related subareas. We will rewrite the connectomics section to make it more accessible, reflecting the change in the figures.

Additionally, identifying the primary inputs to A13 associated with motor function would enhance the study's clarity and relevance.

This is a great point and could help simplify the whole-brain results. We can present the motor-related inputs and outputs as part of a new figure in the main paper and add accompanying text in the results section. This will help highlight possible therapeutic pathways. We can also enhance our discussion of these motor-related pathways. We will retain the entire dataset and present it in a supplementary table for those who are interested.

The study raises intriguing questions about compensatory mechanisms in Parkinson's disease and a new perspective on the preservation of dopaminergic cells in A13, despite the SNc degeneration, and the plastic changes to input/output matrices. To gain inspiration for a more straightforward reanalysis and discussion of the results, I recommend the authors refer to the paper titled "Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon from the David Kleinfeld laboratory." This could guide the authors in investigating motor pathways across different brain regions.

Thank you for the advice, and as pointed out, Kleinfeld's group had a nice, focused presentation of their data. For the connectomic piece, we can certainly adopt their reporting style, which, as you point out, may highlight key motor-related regions. There are a few ideas here that we can explore further, as mentioned above.

> 1. *Description of locomotor performance. Figure 3 provides valuable data on the locomotor effects of A13 region photoactivation in both control and 6-OHDA mice. However, a more detailed analysis of the changes in locomotion during stimulation would enhance our understanding of the pro-locomotor effects, especially in the context of 6-OHDA lesions. For example, it would be informative to explore whether the probability of locomotion changes during stimulation in the control and 6-OHDA groups. Investigating reaction time, speed, total distance, and even kinematic aspects during stimulation could reveal how A13 is influencing locomotion, particularly after 6-OHDA lesions. The laboratory of Whelan has a deep knowledge of locomotion and the neural circuits driving it so these features may be instructive to infer insights on the neural circuits driving movement. On the same line, examining features like the frequency or power of stimulation related to walking patterns may help elucidate whether A13 is engaging with the Mesencephalic Locomotor Region (MLR) to drive the pro-locomotor effects. These insights would provide a more comprehensive understanding of the mechanisms underlying A13-mediated locomotor changes in both healthy and pathological conditions.*

Thank you for these suggestions. We will revise as suggested. We will provide additional and/or updated data in revised figures and text. We will also move Supplementary Figures S1 and S2, which present additional locomotor data, into the main text to partly address the reviewers' points.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *The paper by Kim et al. investigates the potential of stimulating the dopaminergic A13 region to promote locomotor restoration in a Parkinson's mouse model. Using wild-type mice, 6-OHDA injection depletes dopaminergic neurons in the substantia nigra pars compacta, without impairing those of the A13 region and the ventral tegmentum area, as previously reported in humans. Moreover, photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region improves bradykinesia and akinetic symptoms after 6-OHDA injection. Whole-brain imaging with retrograde and anterograde tracers reveals that the A13 region undergoes substantial changes in the distribution of its afferents and projections after 6-OHDA injection. The study suggests that if the remodeling of the A13 region connectome does not promote recovery following chronic dopaminergic depletion, photostimulation of the A13 region restores locomotor functions.*
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> *Strengths:*
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> *Photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region promotes locomotion and locomotor recovery of wild-type mice 1 month after 6-OHDA injection in the medial forebrain bundle, thus identifying a new potential target for restoring motor functions in Parkinson's disease patients.*
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> *Weaknesses:*

Electrical stimulation of the medial Zona Incerta, in which the A13 region is located, has been previously reported to promote locomotion (Grossman et al., 1958). Recent mouse studies have shown that if optogenetic or chemogenetic stimulation of GABAergic neurons of the

Zona Incerta promotes and restores locomotor functions after 6-OHDA injection (Chen et al., 2023), stimulation of glutamatergic ZI neurons worsens motor symptoms after 6-OHDA (Lie et al., 2022).

Thank you - we will add this reference. It is useful as Grossman did stimulate the zona incerta in the cat and elicit locomotion, suggesting that stimulation of the area in normal mice has external validity. The area targeted by Chen et al. (2023) is in the lateral aspect of central/medial zona incerta, formed by dorsal and ventral zona incerta, which may account for the differing results. Our data were robust for stimulation of the medial aspect of the rostromedial zona incerta. The thigmotactic behaviour that we observed in our work that focused on CamKII neurons has not been observed with chemogenetic, optogenetic activation or with photoinhibition of GABAergic central/medial ZI (Chen et al. 2023).

> *Although CAMKIIa is a marker of presumably excitatory neurons and can be used as an alternative marker of dopaminergic neurons, behavioral results of this study raise questions about the neuronal population targeted in the vicinity of the A13 region. Moreover, if YFP and CHR2-YFP neurons express dopamine (TH) within the A13 region (Fig. 2), there is also a large population of transduced neurons within and outside of the A13 region that do not, thus suggesting the recruitment of other neuronal cell types that could be GABAergic or glutamatergic.*

We found that CamKII transfection of the A13 region was extremely effective in promoting locomotor activity, which was critical for our work in exploring its possible therapeutic potential. We acknowledge that specific viral approaches that target the GABAergic, glutamatergic, and dopaminergic circuits would be very useful. The range of tools to target A13 dopaminergic circuits is more limited than the SNc, for example, because the A13 region lacks DAT, and TH-IRES-Cre approaches, while useful, are less specific than DAT-Cre mouse models. Intersectional approaches targeting multiple transmitters (glutamate & dopamine, for example) may be one solution as we do not expect that a single transmitter-specific pathway would work, as well as broad targeting of the A13 region. Recent work suggests that GABAergic neuron activation may have more general effects on behaviour rather than control of ongoing locomotor parameters. However, this is in contrast to recent work showing a positive valence effect of dopamine A13 activation on motivated food-seeking behavior, which differs from consummatory behavior observed with GABAergic modulation (Ye, Nunez, and Zhang 2023). Chemogenetic inactivation and ablation of dopaminergic A13 revealed that they contribute to grip strength and prehensile movements, uncoupling food-seeking grasping behavior from motivational factors (Garau et al. 2023). Overall, this suggests differing effects of GABA compared to DA and/or glutamatergic cell types, consistent with our effects of stimulating CamKII.

> *Regarding the analysis of interregional connectivity of the A13 region, there is a lack of specificity (the viral approach did not specifically target the A13 region), the number of mice is low for such correlation analyses (2 sham and 3 6-OHDA mice), and there are no statistics comparing 6-OHDA versus sham (Fig. 4) or contra- versus ipsilesional sides (Fig. 5). Moreover, the data are too processed, and the color matrices (Fig. 4) are too packed in the current format to enable proper visualization of the data. The A13 afferents/efferents analysis is based on normalized relative values; absolute values should also be presented to support the claim about their upregulation or downregulation.*

Generally, papers using tissue-clearing imaging approaches have low sample sizes due to technical complexity and challenges. The technical challenges of obtaining these data were substantial in both collection and analysis. There are multiple technical complexities arising from dual injections (A13 and MFB coordinates) and targeting the area correctly. The A13 region is difficult to target as it spans only around 300 μm in the anterior-posterior axis. While clearing the brain takes weeks, and light-sheet imaging also takes time, the time necessary to analyze the tissue using whole-brain quantification is labor intensive, especially with a lack of a standardized analysis pipeline from atlas registrations, signal segmentations, and quantifications. The field is still relatively new, requiring additional time to refine pipelines.

Correlation matrices are often used in analyzing connectivity patterns on a brain-wide scale, as they can identify any observable patterns within a large amount of data. We used correlation matrices to display estimated correlation coefficients between the afferent and efferent proportions from one brain subregion to another across 251 brain regions in total in a pairwise manner (not for hypothesis testing). We provided descriptive statistics (mean and error bars) in Figure 5C and G. As mentioned in comments for Reviewer 1, we will also present data in a revised Figure 5 and/or a new figure that focuses specifically on motor-related pathways to provide information on possible therapeutic pathways. As suggested, absolute values will be shared in a supplemental table.

> *In the absence of changes in the number of dopaminergic A13 neurons after 6-OHDA injection, results from this correlation analysis are difficult to interpret as they might reflect changes from various impaired brain regions independently of the A13 region.*

We acknowledge that models of Parkinson's disease, particularly those using 6-OHDA, induce plasticity in various regions, which may subsequently affect A13 connectivity. Our aim is to emphasize the residual, intact A13 pathways that could serve as therapeutic targets in future investigations. This emphasis is pertinent in the context of potential clinical applications, as the overall input and output to the region fundamentally dictate the significance of the A13 region in lesioned nigrostriatal models. We agree with the reviewer that the changes certainly can be independent of A13; however, the fact that there was a significant change in the connectome post-6-OHDA injection and striatonigral degeneration is in and of itself important and important to document.

> *There is no causal link between anatomical and behavioral data, which raises questions about the relevance of the anatomical data.*

This point was also addressed earlier in response to a comment from Reviewer 1. Focusing on specific motor pathways is one avenue to explore. However, given that the zona incerta acts as an integrative hub, we believed it is prudent to initially examine both afferent and efferent pathways using a brain-wide approach. For instance, without employing this methodology, the potential significance of cortical interconnectivity to the A13 region might not have been fully appreciated. As mentioned previously, we will place additional emphasis on motor-related regions in our revised paper, thereby enhancing the relevance of the anatomical data presented. With these modifications, we anticipate that our data will underscore specific motor-related targets for future exploration, employing optogenetic targeting to assess necessity and sufficiency.

> *Overall, the study does not take advantage of genetic tools accessible in the mouse to address the direct or indirect behavioral and anatomical contributions of the A13 region to motor control and recovery after 6-OHDA injection.*

We acknowledge that our study has not specifically targeted neurons that express dopaminergic, glutamatergic, or GABAergic properties (refer to earlier comment for more detail). However, like others, we find that targeting one neuronal population often does not result in a pure transmitter phenotype. For instance, evidence suggests co-localization of dopamine neurons with a subpopulation of GABA neurons in the A13/medial zona incerta

(Negishi et al. 2020). In the hypothalamus, research by Deisseroth and colleagues (Romanov et al. 2017) indicates the presence of multiple classes of dopamine cells, each containing different ratios of co-localized peptides and/or fast neurotransmitters. Consequently, we believe our work lays the foundation for the investigations suggested by the reviewer. Furthermore, if one considers this work in the context of a preclinical study to determine whether the A13 might be a target in human Parkinson's disease, the existing technology that could be utilized is deep brain stimulation (DBS) or electrical modulation, which would also affect different neuronal populations in a non-specific manner. While optogenetic stimulation therapy is longer term, using CamKII combined with the DJ hybrid AAV could be a translatable strategy for targeting A13 neuronal populations in non-human primates (Watakabe et al. 2015; Watanabe et al. 2020).

> ***Reviewer #3 (Public Review):***
>
> *Kim, Lognon et al. present an important finding on pro-locomotor effects of optogenetic activation of the A13 region, which they identify as a dopamine-containing area of the medial zona incerta that undergoes profound remodeling in terms of afferent and efferent connectivity after administration of 6-OHDA to the MFB. The authors claim to address a model of PD-related gait dysfunction, a contentious problem that can be difficult to treat with dopaminergic medication or DBS in conventional targets. They make use of an impressive array of technologies to gain insight into the role of A13 remodeling in the 6-OHDA model of PD. The evidence provided is solid and the paper is well written, but there are several general issues that reduce the value of the paper in its current form, and a number of specific, more minor ones. Also, some suggestions, that may improve the paper compared to its recent form, come to mind.*

Thank you for the suggestions and careful consideration of our work - it is appreciated.

> *The most fundamental issue that needs to be addressed is the relation of the structural to the behavioral findings. It would be very interesting to see whether the structural heterogeneity in afferent/effects projections induced by 6-OHDA is related to the degree of symptom severity and motor improvement during A13 stimulation.*

As mentioned in comments for Reviewer 1, we will be highlighting motor-related A13 pathways in a revised Figure 5 and/or a new figure. We hope that our work will provide a roadmap for future studies to disentangle divergent or convergent A13 pathways that are involved in different or all PD-related motor symptoms. Because we could not measure behavioural change in the same animals studied with the anatomic study (essentially because the optrode would have significantly disrupted the connectome we are measuring), we cannot directly compare behaviour to structure.

> *The authors provide extensive interrogation of large-scale changes in the organization of the A13 region afferent and efferent distributions. It remains unclear how many animals were included to produce Fig 4 and 5. Fig S5 suggests that only 3 animals were used, is that correct? Please provide details about the heterogeneity between animals. Please provide a table detailing how many animals were used for which experiment. Were the same animals used for several experiments?*

The behavioral set and the anatomical set were necessarily distinct. In the anatomical experiments, we employed both anterograde and retrograde viral approaches to target the afferent and efferent A13 populations with fluorescent proteins. For the behavioral approach, a single ChR2 opsin was utilized to photostimulate the A13 region; hence combining the two populations was not feasible. We were also concerned that the optrode itself would interfere with connectomics. A lower number of animals were used for the whole-brain work due to technical limitations described earlier. We will provide more details regarding numbers we can identify as a table and text.

> *While the authors provide evidence that photoactivation of the A13 is sufficient in driving locomotion in the OFT, this pro-locomotor effect seems to be independent of 6-OHDA-induced pathophysiology. Only in the pole test do they find that there seems to be a difference between Sham vs 6-OHDA concerning the effects of photoactivation of the A13. Because of these behavioral findings, optogenic activation of A13 may represent a gain of function rather than disease-specific rescue. This needs to be highlighted more explicitly in the title, abstract, and conclusion.*

We agree with the reviewer that this aspect needs to be highlighted more. Optogenetic activation of A13 may represent a gain of function in both healthy and 6-OHDA mice, highlighting a parallel descending motor pathway that remains intact. 6-OHDA lesions have multiple effects on motor and cognitive function. This makes a single pathway unlikely to rescue all deficits observed in 6-OHDA models. We can say that the lack of locomotion observed in 6-OHDA models can be reversed by A13 region stimulation. We have discussed some aspects of the gain of function possibility but will augment this in other areas of the paper as well, as suggested.

> *The authors claim that A13 may be a possible target for DBS to treat gait dysfunction. However, the experimental evidence provided (in particular the lack of disease-specific changes in the OFT) seems insufficient to draw such conclusions. It needs to be highlighted that optogenetic activation does not necessarily have the same effects as DBS (see the recent review from Neumann et al. in Brain: https://pubmed.ncbi.nlm.nih.gov/37450573/). This is important because ZI-DBS so far had very mixed clinical effects. The authors should provide plausible reasons for these discrepancies. Is cell-specificity, which only optogenetic interventions can achieve, necessary? Can new forms of cyclic burst DBS achieve similar specificity (Spix et al, Science 2021)? Please comment.*

Thank you for the useful comments - we will update our discussion accordingly.

Our study highlights a parallel motor pathway provided by the A13 region that remains intact in 6-OHDA mice and can be sufficiently driven to rescue the hypolocomotor pathology observed in the OFT and overcome bradykinesia and akinesia. The photoactivation of ipsilesional A13 also has an overall additive effect on ipsiversive circling, representing a gain of function on the intact side that contributes to the magnitude of overall motor asymmetry against the lesioned side. The effects of DBS are rather complex, ranging from micro-, meso-, to macro-scales, involving activation, inhibition, and informational lesioning, and network interactions. This could contribute to the mixed clinical effects observed with ZI-DBS, in addition to differences in targeting and DBS programming among the studies (see review (Ossowska 2019)). Also the DBS studies targeting ZI have never targeted the rostromedial ZI which extends towards the hypothalamus and contains the A13. Furthermore, DBS and electrical stimulation of neural tissue, in general, are always limited by current spread and lower thresholds of activation of axons (e.g., axons of passage), both of which can reduce the specificity of the true therapeutic target. Optogenetic studies have provided mechanistic insights that could be leveraged in overcoming some of the limitations in targeting with conventional DBS approaches. Spix et al. (2021) provided an interesting approach highlighting these advancements. They devised burst stimulation to facilitate population-specific neuromodulation within the external globus pallidus. Moreover, they found a complementary role for optogenetics in exploring the pathway-specific activation of neurons activated by DBS. To ascertain whether A13 DBS may be a viable therapy for PD gait, it will be necessary to perform many more preclinical experiments, and tuning of DBS parameters could be facilitated by optogenetic stimulation in these murine models.

In a recent study, Jeon et al (Topographic connectivity and cellular profiling reveal detailed input pathways and functionally distinct cell types in the subthalamic nucleus, 2022, Cell Reports) provided evidence on the topographically graded organization of STN afferents and McElvain et al. (Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon, 2021, Neuron) have shown similar topographical resolution for SNr efferents. Can a similar topographical organization of efferents and afferents be derived for the A13/ ZI in total?

The ZI can be subdivided into four subregions in the antero-posterior axis: rostral (ZIr), dorsal (ZId), ventral (ZIv), and caudal (ZIc) regions. The dorsal and ventral ZI is also referred together as central/medial/intermediate ZI. There are topographical gradients in different cell types and connectivity across these subregions (see reviews: (Mitrofanis 2005; Monosov et al. 2022; Ossowska 2019). Recent work by Yang and colleagues (2022) demonstrated a topographical organization among the inputs and outputs of GABAergic (VGAT) populations across four ZI subregions. Given that A13 region encompasses a smaller portion (the medial aspect) of both rostral and medial/central ZI (three of four ZI subregions) and coexpress VGAT, A13 region likely falls under rostral and intermediate medial ZI dataset found in Yang et al. (2022). With our data, we would not be able to capture the breadth of topographical organization shown in Yang et al (2022).

In conclusion, this is an interesting study that can be improved by taking into consideration the points mentioned above.

Reviewer #1 (Recommendations For The Authors):

1. *Figure 2 indeed presents valuable information regarding the effects of A13 region photoactivation. To enhance the comprehensiveness of this figure and gain a deeper understanding of the neurons driving the pro-locomotor effect of stimulation, it would be beneficial to include quantifications of various cell types:*

   • cFos-Positive Cells/TH-Positive Cells: it can help determine the impact of A13 stimulation on dopaminergic neurons and the associated pro-locomotor effect in the healthy condition and especially in the context of Parkinson's disease (PD) modeling.

   • cFos-Positive Cells /TH-Negative Cells: Investigating the number of TH-negative cells activated by stimulation is also important, as it may reveal non-dopaminergic neurons that play a role in locomotor responses. Identifying the location and characteristics of these TH-negative cells can provide insights into their functional significance.

   Incorporating these quantifications into Figure 2 would enhance the figure's informativeness and provide a more comprehensive view of the neuronal populations involved in the locomotor effects of A13 stimulation.

Agreed - we will add quantification and create graphs to present the data in Figure 2.

1. *Refer to Figure 3. In the main text (page 5) when describing the animal with 6-OHDA the wrong panels are indicated. It is indicated in Fgure 2A-E but it should be replaced with 3A-E. Please do that.*

Will be done

Reviewer #2 (Recommendations For The Authors):

Abstract

> *Page 1: Inhibitory or lesion studies will be necessary to support the claim that the global remodeling of afferent and efferent projections of the A13 region highlights the Zona Incerta's role as a crucial hub for the rapid selection of motor function.*

We believe that overall, there is quite a bit of evidence that the zona incerta is a hub for afferent/efferents. Mitrofanis (2005) and, more recently, Wang et al. (2020) summarize some of the evidence. Yang (2022) illustrates that the zona incerta shows multiple inputs to GABAergic neurons and outputs to diverse regions. Recent work suggests that the zona incerta contributes to various motor functions such as hunting, exploratory locomotion, and integrating multiple modalities (Zhao et al. 2019; Wang et al. 2019; Monosov et al. 2022; Chometton et al. 2017). We will update our paper to reflect these references.

> *Introduction*
>
> *Page 2, paragraph 2: "However, little attention has been placed on the medial zona incerta (mZI), particularly the A13, the only dopamine-containing region of the rostral ZI" Is the A13 region located in the rostral or medial ZI or both?*

It should have been written "rostromedial" ZI. The A13 is located in the medial aspect of rostromedial ZI. We will update the introduction.

> *Page 2, para 3: Li et al (2021) used a mini-endoscope to record the GCaMP6 signal. Masini and Kiehn, 2022 transiently blocked the dopaminergic transmission; they never used 6-OHDA. Please correct through the text.*

We will correct this.

> *Page 2, para 4: the A13 connectome encompasses the cerebral cortex,... MLR. The MLR is a functional region, correct this for the CNF and PPN.*

Thank you, we will correct this.

> *Page 3, the last paragraph of the introduction could be clarified by presenting the behavioral data first, followed by the anatomy.*

We will correct this.

> *Figure 1 is nice and clear, and well summarizes the experimental design.*

Thank you.

> *Figure 2 shows an example of the extent of the ChR2-YFP expression and the position of an optical fiber tip above the dopaminergic A13 region from a mouse. Without any quantification, these images could be included in Figure 1. Despite a very small volume (36.8nL) of AAV, the extent of ChR2-YFP expression is quite large and includes dopaminergic and unidentified neurons within the A13 region but also a large population of unidentified neurons outside of it, thus raising questions about the volume and the types of neurons recruited.*

This is an important consideration. As mentioned previously, we will provide more information on viral spread and optrode location. The issue of viral spread is complex and depends on factors including tissue type, serotype, and promotor of the virus. Li et al. (2021), for example, used different virus serotypes and promotors, injecting 150 nL, whereas we used AAV DJ, injecting 36.8nL. AAV-DJ is a hybrid viral type consisting of multiple serotypes. It has a high transduction efficiency, which leads to greater gene delivery than single-serotype AAV viral constructs (Mao et al. 2016). A secondary consideration regarding translation was that AAV-DJ could effectively transduce non-primate neurons (Watanabe et al. 2020). We have addressed the issue of neurons recruited earlier and will provide c-Fos quantification to illustrate the extent of co-localization with TH.

> *Anatomical reconstruction of the extent of the ChR2-YFP expression and the location of the tip of the optical fiber will be necessary to confirm that ChR2-YFP expression was restricted to the A13 region.*

We will provide additional information regarding viral spread, ferrule tip placement, and c-fos cell counts.

> *Page 5, 1st para: Double-check the references, as not all of them are 6-OHDA injections in the MLF.*

Will correct.

> *Page 5, 1st para, 4th line: Replace ferrule with optical canula or fiber.*

Will correct.

> *Page 5, 1st para, 9th line: Replace Figure 2 with Figure 3.*

Will correct.

> *Page 5, 2nd para: About the refractory decrease in traveled distance by sham-ChR2 mice: is this significant?*

It was not significant (Figure S1, 1-way RM ANOVA: $F_{5,25} = 0.486$, $P = 0.783$)). We will update this.

> *Figure 3 showing behavioral assessments is nice, but the stats are not always clear. In Fig 3A, are each of the off and on boxes 1 minute long? The figure legend states the test lasts 1 min, but isn't it 4 minutes? In Figure 3B-E and 3J-M, what are the differences? Do the stats identify a significant difference only during the stimulation phase? Fig. 3F-I are nice and could have been presented as primary examples prior to data analysis in Fig. 3B-E. Group labels above the graph would help.*

Yes, the off-on boxes are 1 minute long. We will correct the error in the legend. Great suggestion for F-I - we will move them ahead of the summary figures.

> *Fig. 3L-M, what do PreSur, Post, and Ferrule mean? I assume that Ferrule refers to mice tested with the optical fiber without stimulation, whereas Stim. refers to the stimulation. It would be helpful to standardize the format of stats in Fig. 3B-E and 3-J-M. What are time points a, b, and c referring to?*

We will do this.

> *Figure S2A: the higher variability in 6-OHDA-YFP mice in comparison to 6-OHDA-ChR2 mice prior to stimulation suggests that 6-OHDA-YFP mice were less impaired. Why use boxplots only for these data? Would a pairwise comparison be more appropriate?*

Data did not follow a normal distribution and thus, were plotted as box and whiskers with the horizontal line through the box indicating the group median, interquartile range indicated by the limits of the box, and group minimum and maximum indicated by the whiskers. And indeed, a non-parametric equivalent of paired t-test (Wilcoxon signed-rank test) was used.

> *Fig. S2B: add the statistical marker.*

Will do

> *Page 7, para 1, line 8: to add "in comparison to 6-OHDA-YFP and YFP mice" to during photostimulation... (Figure 3E).*

Will do

> *Page 7, para 3, line 5: about larger improvement, replace "sham ChR2" with "6-OHDA."*

Will do

> *Page 8, para 1, line 4: Perier et al., 2000 reported that 6-OHDA injection increased the firing frequency of the ZI over a month.*

We will add that time frame. Agreed, it is shorter than the behavioral work, which was started 3 weeks after 6-OHDA injection.

> *Page 8, para 2, line 1: Since the results were expected, add some references.*

Will do

> *Page 8, para 3, line 4. Double-check the reference.*

Will correct and update

> *Page 8: About large-scale changes in the A13 region, the relevance of correlation matrices is difficult to grasp. Analysis of local connectivity would have been more informative in the context of GABAergic and glutamatergic neurons of the ZI in the vicinity of the A13 region.*

We will explore alternative methods to present the data.

> *Page 8, para 3, line: given Fig. 2, there is concern about the claim that only the A13 region was targeted. The time of the analysis after 6-OHDA should be mentioned. Some sections of the paragraph could be moved to methods.*

As mentioned earlier, we will provide additional information regarding viral spread, ferrule tip placement, and c-fos cell counts. We will mention analysis time after 6-OHDA and update Figure 1a to include this.

> *Fig. 4: The color code helps the reader visualize distribution differences. However, statistical analyses comparing 6-OHDA versus sham should be included. Quantification per region would greatly help readers visualize the data and support the conclusion. The relationship between the type of correlation (positive or negative) and absolute change (increase or decrease) is unknown in the current format, which limits the interpretation*

> *of the data. Moreover, examples of raw images of axons and cells should be presented for several brain regions. The experimental design with a timeline, as in Fig. 1, would be helpful. The legend for Fig. 4 is a bit long. Some sections are very descriptive, whereas others are more interpretive.*

We will explore alternative methods of presenting the data, as suggested in a previous comment. Should we retain the correlation matrix, we will incorporate the reviewer's suggestions.

> *Page 10, para 1, line 1: add "afferent" to "changes in -afferent and- projection patterns."*

Will do

> *Page 10, para 1, line 9: remove the 2nd "compared to sham" in the sentence.*

Will do

> *10, para 1, line 10: remove "coordinated" in "several regions showed a coordinated reduction in afferent density." We cannot say anything about the timing of events, as there is only info at 1 month.*

Will do

> *Page 10, para 2: the section should be written in the past tense.*

Will do

> *Page 13, para 2, the last sentence is overstated. Please remove "cells" and refer to the A13 region instead.*

Will do

> *About differential remodelling of the A13 region connectome: Figure 5C and 5G: The proportion of total afferents ipsi- and contralateral to 6-OHDA injection argues that the A13 region primarily receives inputs from the cortical plate and the striatum. Unfortunately, there are no statistics.*

Due to the small sample size, we provided descriptive statistics (mean and error bars) in Figure 5C and G. As mentioned in comments for Reviewers 1 and 2, we will revise Figure 5 to present data focusing on motor-related pathways to provide clarity. In addition, absolute values will be shared in a supplemental table.

> *Figure 5 D and 5H: Changes in the proportion of total afferents/projections are relatively modest (less than 10% of the whole population for the highest changes). There is no standard deviation for these data and no statistics. Do they reflect real changes or variability from the injection site?*

The changes are relatively modest (less than 10%) since a small brain region usually provides a very small proportion of total input (McElvain et al. 2021; Yang et al. 2022). The changes in the proportions reflect real differences between average proportions observed in sham and 6-OHDA mice. The variability in the total labeling of neurons and fibers was minimized by normalizing individual regional counts against total counts found in each individual animal.

> *Fig 5F and H: The example in F shows a huge decrease in the striatum, but H indicates only a 2% change, which makes the example not very representative. Absolute values would be helpful.*

While a 2% change may seem small, it represents a relatively large change in the A13 efferent connectome. To provide further clarity, we will provide absolute values as suggested in our new supplemental table.

> *Figure 6 is inaccurate and unnecessary.*

Agree - it is too simplistic. We will remove it and replace it with one outlined in comments to Reviewer 1.

> *Discussion*
>
> *Although interesting, the discussion is too long.*

We will make it more concise in the revised paper.

> *Page 12: para 2. If the A13 region has a pro-locomotor effect and has therapeutical potential; the claim about its plasticity relies on Fig. 4 and 5, which have a limited scope in the current analysis and presentation (see comments above).*

We will revise the paper per the comments above and then update this accordingly.

> *Methods*
>
> *Page 17, para 1: include the stereotaxic coordinates of the optical cannula above the A13 region.*

We will include this information.

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